Catching Bad Surrogate Models Without the Answer

The anchor-symmetry error certificate on seven datasets

Brandon Rodriguez · 16 October 2026

Question: a model is used beyond the data it was trained on. How wrong is it there?

Answer: physics + the machine's published limit say where the truth must be. If the model lands far outside that range, it is wrong by **at least** that distance.

Test: 7 datasets. Real high-range data hidden, then used to check.

Dataset	Straight law: range right	With published limit: range right	Bound valid	Error proven
Kelmarsh wind	3%	100%	100%	0.80
Penmanshiel wind	82%	100%	100%	0.62
Hill of Towie wind	65%	100%	100%	0.72
La Haute Borne wind	100%	100%	100%	0.28
Solar 2107	95%	98%	100%	0.53
Solar 9068	99%	99%	100%	0.02
Capillary waves	100%	100%	100%	0.52

Beyond the published limit. "Error proven" = median share of the model's real error that the certificate guarantees.

- **Never over-claimed** when its range was right: 0 cases.
- **Published limits fixed the wind farms:** 3–82% → 99.7–100%.
- **Ranks 100 models** correctly on 94–99% of pairs (wind, solar 2107).
- **K-S3 confirmed on Kelmarsh 2021:** allowance 14× → 1.6× the real error, still 96% right.
- **Neural PDE surrogates:** never over-claimed; Lipschitz baseline never says anything.

1. The certificate

Three terms. The defect minus two allowances = the guaranteed minimum error.

Physical law with the published limit

$$c(x) = \min\left(\bar{y}(a^*) + q \ln \frac{x}{x_{\text{law}}^{\max}},\ \ln y_{\text{cap}}\right)$$

x	$x_{\text{law}}^{\max}$	$\bar{y}(a^*)$	q	y_{cap}	ln
New input	Calibration ceiling	Measured output at ceiling	Fitted law exponent	Published limit (spec sheet)	Natural logarithm

The power law from the calibration data, stopped at the machine's published limit.

Defect

$$D_\theta(x, \lambda) = \| f_\theta(T_\lambda x) - A_\lambda f_\theta(x) - c_\lambda \|_2$$

x	$T_\lambda x$	f_θ	A_λ	c_λ	$\|\cdot\|_2$
New input	Anchor input (top training bin)	Surrogate model	Law scaling (= I here)	Law shift $c(T_\lambda x) - c(x)$	Euclidean length

The defect is the mismatch between the surrogate and the physical law.

Anchor-error allowance

$$\| f_\theta(T_\lambda x) - f(T_\lambda x) \|_2 \le b_\lambda$$

$T_\lambda x$	$f_\theta(T_\lambda x)$	$f(T_\lambda x)$	b_λ	$\|\cdot\|_2$
Anchor input	Anchor prediction	True (measured) anchor output	Anchor error bound	Euclidean length

Measured at the top training bin, so here b is the surrogate's actual error there.

Law allowance

$$\beta_\lambda(x) = \kappa' \min\left(\left[\ln \frac{x}{x_{\text{law}}^{\max}} \right]_+,\ \delta^* \right) + r_\lambda$$

$$\kappa = \max_h \frac{\|\bar{y}(n) - \hat{c}_h(n)\|_2}{d_h}, \qquad \delta^* = \max_j \frac{\ln y_{\text{cap},j} - \bar{y}_j(a^*)}{q_j}, \qquad r_\lambda = \|f(T_\lambda x) - c(T_\lambda x)\|_2$$

$\kappa' = 2\kappa$	$[\cdot]_+$	δ^*	r_λ	$\beta_\lambda(x)$
Declared growth rate	Clip negatives to zero	Distance where law hits cap	Law's measured miss at anchor	Law departure bound

Grows with the log-distance past the calibration ceiling, and stops growing at the published limit.

Full certificate

$$\|f_\theta(x) - f(x)\|_2 \geq S(x) = \max_{\lambda \in \Lambda_{\text{anc}}(x)} \frac{\left[\sum_{j=1}^{3} r_j(x, \lambda) \right]_+}{\|A_\lambda\|_2}$$

$r_1 = D_\theta(x, \lambda)$	$r_2 = -b_\lambda$	$r_3 = -\beta_\lambda(x)$	$[u]_+ = \max(u, 0)$	$\Lambda_{\text{anc}}(x)$
Defect	Minus anchor allowance	Minus law allowance	Clip at zero	Admitted anchors (one here)

The guaranteed minimum error: the defect minus both allowances, if both allowances hold.

Symbol	In these tests
$T_\lambda x$ (anchor)	top training bin (6.25 m/s · 325 W/m² · 54 mW)
$x^{\max}_{\text{law}}$ (ceiling)	top calibration bin (8.25 m/s · 525 W/m² · 146 or 192 mW)
A_λ	identity: outputs are logarithms
κ'	2κ, the safety factor 2 on the worst calibration growth rate

Example. Turbine rated 2,050 kW, wind 13 m/s. Allowed range ≈ 2,050 kW ± 22%. A model predicting 9,000 kW is wrong by **at least ×3.5**. A model predicting 2,000 kW: S = 0 (no proven error).

2. Methods for the law allowance β

β is how wrong the physical law itself may be. Every method below is a different way to set it.

Which method to use

Your system	Use	Why
Published limit **and** earlier years of the same machines	**K-S3**	Spec gives the bend; history gives the plateau scatter
Output has a **published limit** (rated power, inverter rating)	**K-S1**	The spec gives the bend the data cannot show
No limit, but **many similar units** measured in the far range	**E3**	Other units show the far-range error, with a statistical guarantee
No limit; the law stays **smooth** beyond the data	**E1** (= L)	Error keeps growing at the measured rate; ×2 covers it
Output can only **flatten**, no spec	**Z**	Shape knowledge caps the error
Unknown bend, one system, no spec	**none**	No calibration data can reveal it: add theory, simulation, or data

Every method at a glance

Method	Formula	Needs	Works when	Our result (range right)	Why it works
C theory distance	$\|q - q_{theory}\| \cdot \delta$	theory exponent	never reliably	wind 0%, solar 25%	none: a good fit shrinks the band
P short hops	$2\kappa'\delta$, κ' from hops inside calibration	≥ 3 calibration bins	error grows linearly at the short-hop rate	wind 52%, solar 90%	Lipschitz extrapolation [3, 4]
E1 long hops	$2\kappa\delta$, κ from hops over train + calibration	≥ 8 measured bins	error grows at most linearly	solar 95.5%; wind 68–72% below rated	Lipschitz extrapolation [3, 4]
E2 + curvature	max(E1, 2 × quadratic − line)	same as E1	the bend is inside the data	no gain on wind	Taylor remainder: sees only measured curvature
A offset + slope	$r_0 + 2\kappa_A\delta$	same as E1	a fixed offset plus linear growth	worse everywhere	—
Z saturation	curve may only flatten	shape knowledge	output saturates (concave)	wind 100%, solar 85%	shape-constrained regression [5]
E3 fleet conformal	90% quantile of other units' law error at that input	≥ 9 comparable units	units are exchangeable	wind 96% / 99.5%; solar years 77%	split / Mondrian conformal [6, 7, 8]
F-NN, F-KAN, F-GBT	learned error scale × conformal quantile	fleet data	units are exchangeable	valid 95–100%; no tighter than trees	CV+ [9]; KAN [10]
Per-output	one β per output	many hops per output	enough data for each output	coverage −14 to −21 pts	—
K-L capped law	law capped at the spec; $\beta = 2\kappa\delta$	published output limit	machine reaches its rating	valid; proves 44–65%	turbine and inverter control [11–14]
K-S1 capped + frozen	$\beta = 2\kappa \cdot \min(\delta, \delta^*)$	published limit for each capped output	rating reached; plateau flat	93–100%; proves 62–85%	turbine and inverter control [11–14]
K-S3 capped + shrink	past the cap: min($2\kappa\delta^*$, fleet q_{90})	published limit + earlier years	plateau behaves like past years	2021: 96%; proves 93%	control [11–14] + Mondrian conformal [6]
K-S2 early freeze	freeze at the first cap	—	never: freezes too early	29%	—

δ = log-distance past the calibration ceiling. Full references at the end.

Why each family works

- **Rate × distance (P, E1, K-L, K-S1).** If the law's error changes no faster than a rate κ, it is at most κ × distance away (a Lipschitz bound) [3, 4]. The factor 2 covers a rate measured on short or noisy hops.
- **Published limits (K-L, K-S1).** Turbines hold power at rated by pitching the blades and cap rotor speed; inverters clip at their AC rating [11–14]. So the law is known to go flat, and its error stops growing.
- **Conformal (E3, F-*).** If the new unit behaves like the calibration units, a 90% quantile of their errors covers its error with ≥ 90% probability, with no model assumptions [6–9].
- **Shape (Z).** If the curve can only flatten, its possible departure is limited [5].
- **The bound itself.** If the truth is inside a known region, the distance to the region is a guaranteed minimum error: a distance-to-admissible-set bound, as in set-membership identification [1, 2].

3. Kelmarsh wind 2020

Six Senvion MM92 turbines (UK). 72 turbine-months. Power + rotor speed. Confirmation year.

Law for this dataset

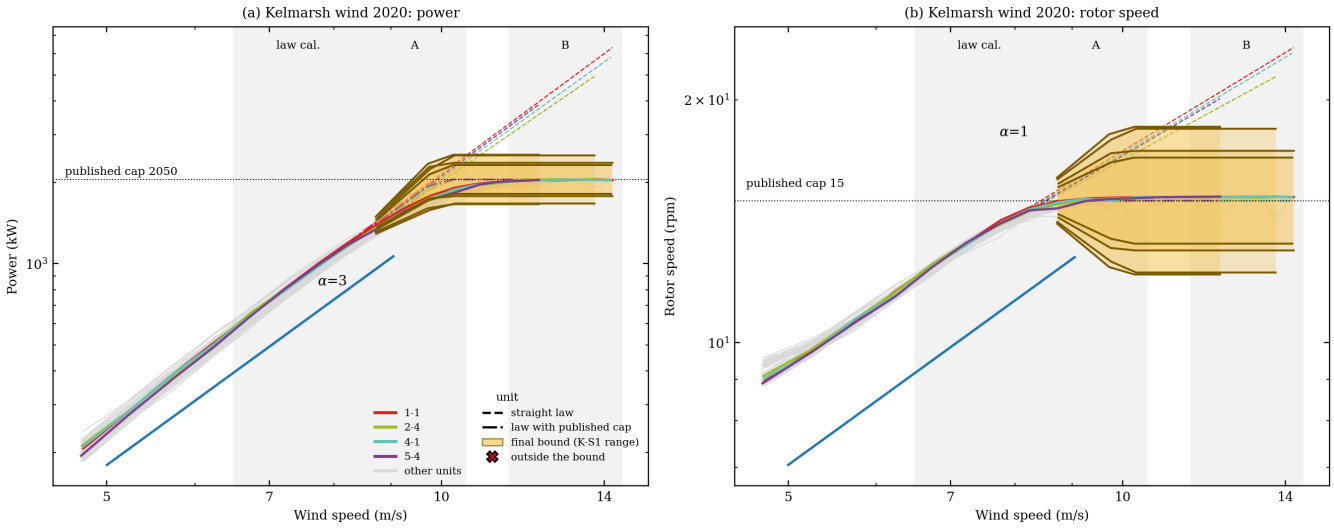
$$c_P(U) = \min\left(\bar{y}_P(a^*) + q_P \ln \frac{U}{U_{\text{law}}^{\max}},\ \ln P_{\text{rated}}\right), \quad c_\omega(U) = \min\left(\bar{y}_\omega(a^*) + q_\omega \ln \frac{U}{U_{\text{law}}^{\max}},\ \ln \omega_{\max}\right)$$

U	$U_{\text{law}}^{\max}$	q_P, q_ω	P_{rated}	$\omega_{\max}$
Wind speed (m/s)	8.5 m/s band top	Fitted (theory 3 and 1)	2050 kW	15.0 rpm

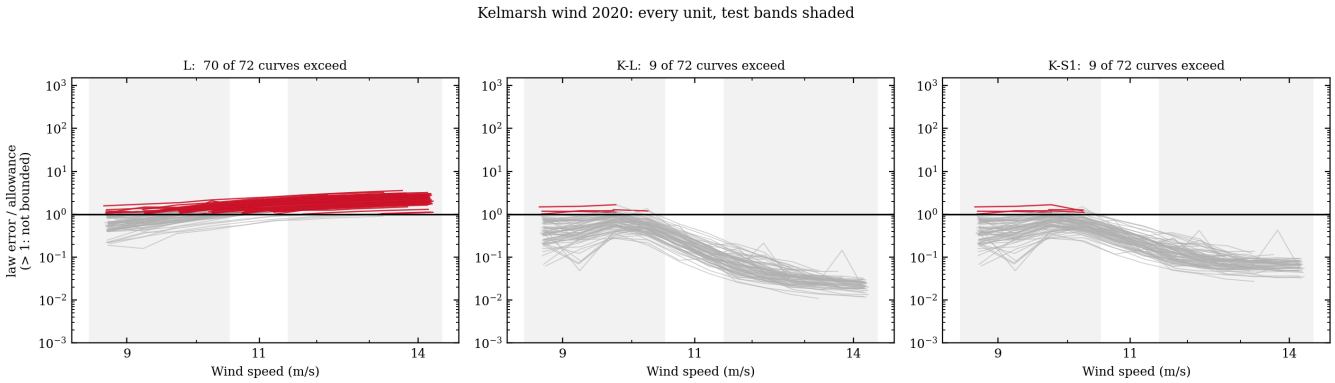
Power and rotor speed follow the law until their published limits (Senvion MM92).

Test band	Rule	Range right	Bound valid	Error proven
below limit	L	67%	92%	0.88
below limit	K-L	94%	98%	0.84
below limit	K-S1	93%	98%	0.85
beyond limit	L	3%	22%	1.19
beyond limit	K-L	100%	100%	0.60
beyond limit	K-S1	100%	100%	0.80

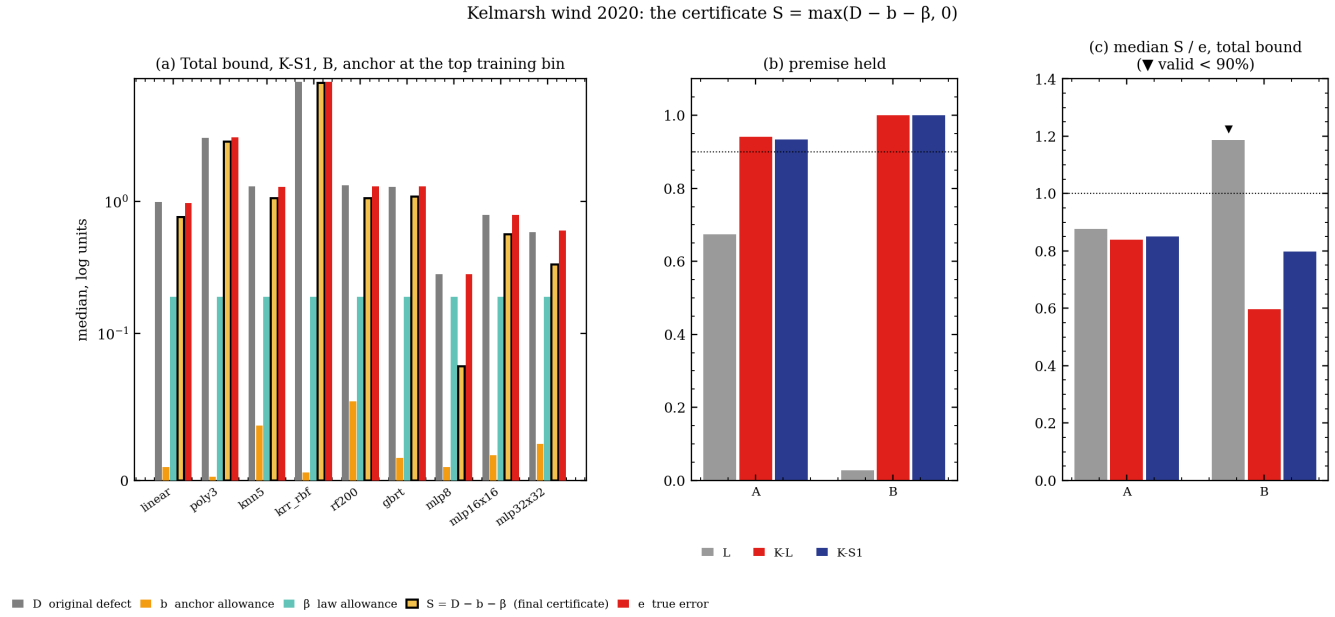
- **Straight law: 3% range right beyond rated → K-S1: 100%.**
- **Proves 80% of the error beyond rated.**



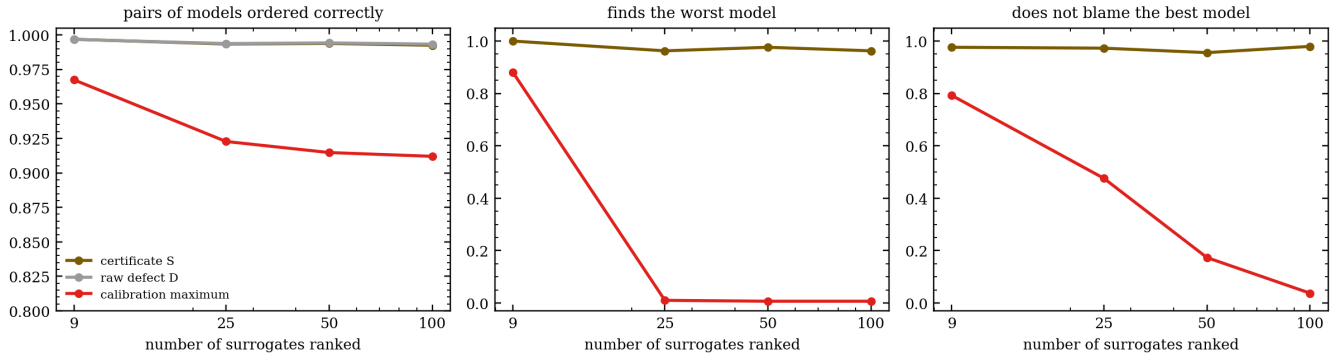
Gold band = allowed range. Red × = point outside it.



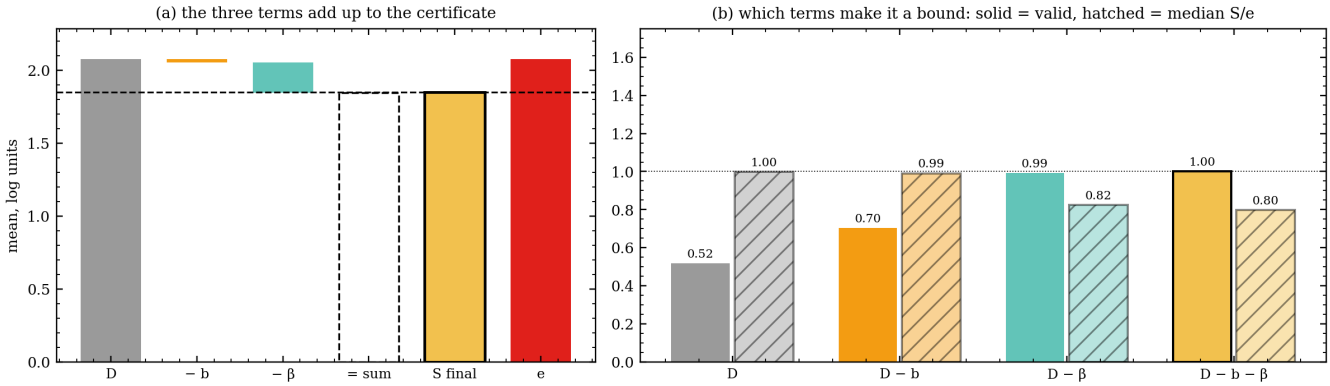
Law error ÷ allowance for every unit. Above 1 = range wrong.



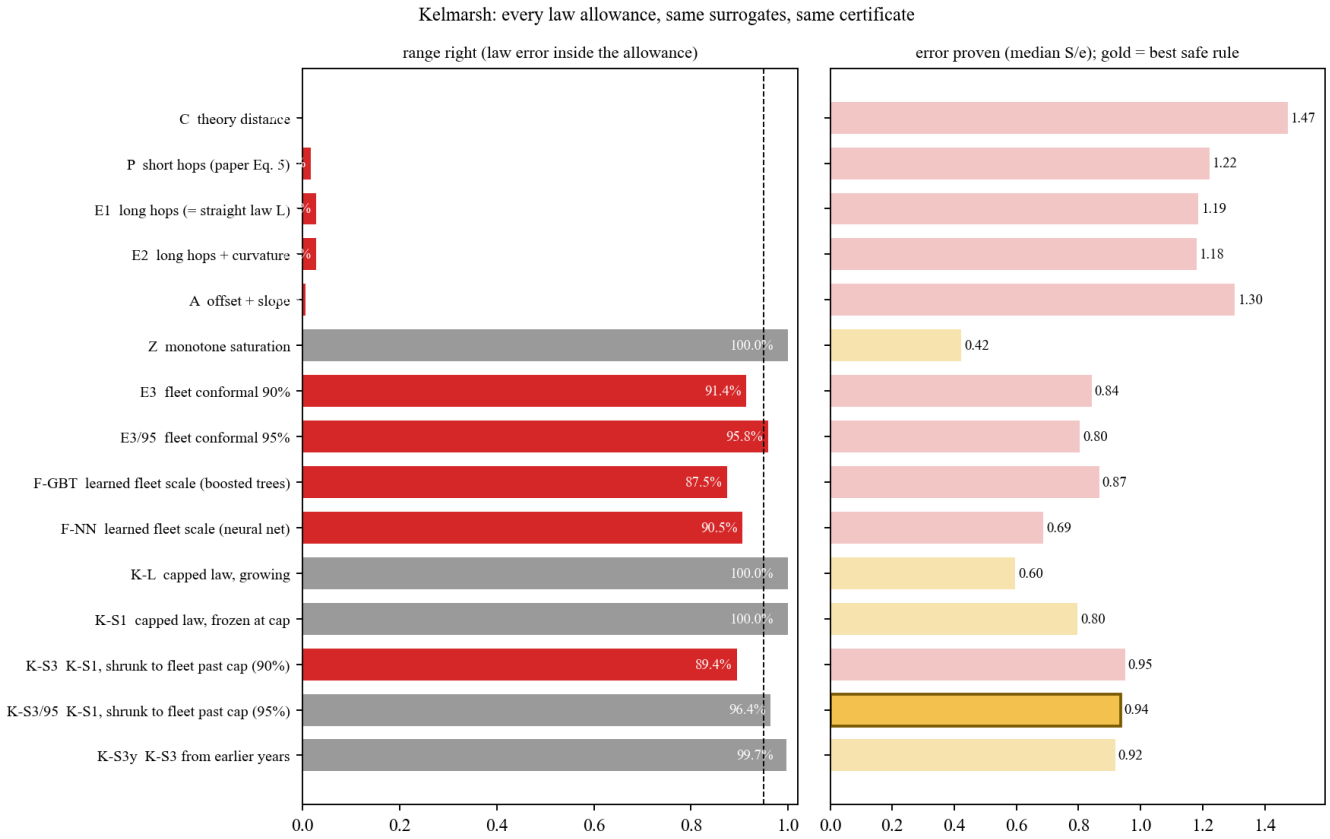
Gold = the certificate S. Red = true error.



Ranking 9 → 100 models: certificate (gold) vs simple baseline (red).



Ablation: $D - b - \beta$ adds up to the certificate (gold).



All 13 law allowances on this dataset (section 11). Red = unsafe; gold = best safe rule.

4. Penmanshiel wind 2017

New. Five Senvion MM82 turbines (Scotland). 60 turbine-months. Power + rotor speed.

Law for this dataset

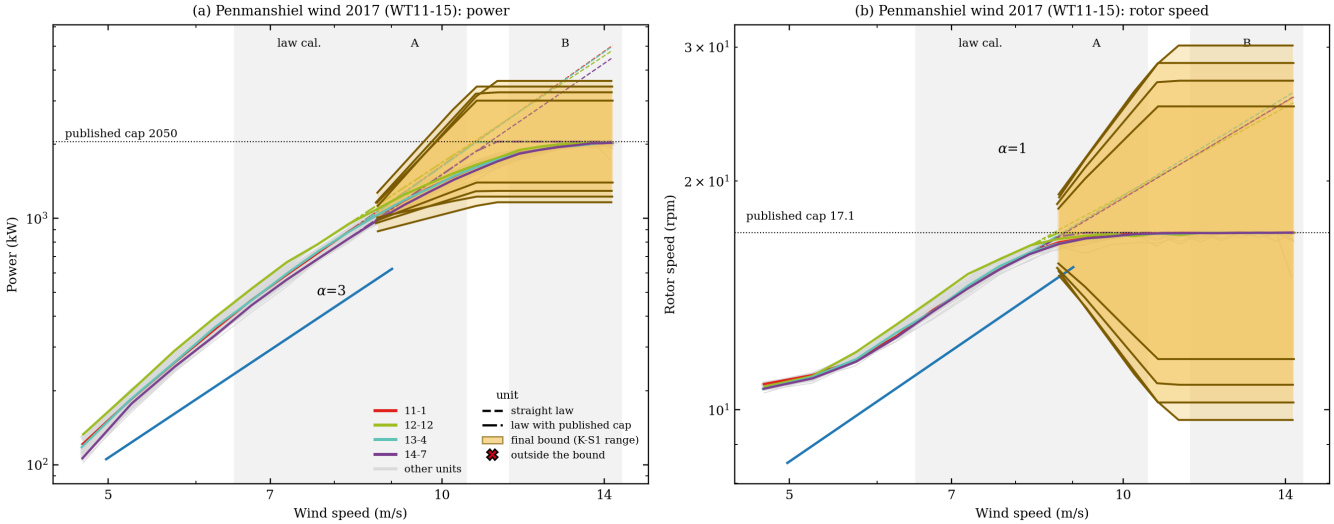
$$c_P(U) = \min\left(\bar{y}_P(a^*) + q_P \ln\frac{U}{U_{\text{law}}^{\max}},\ \ln P_{\text{rated}}\right), \quad c_\omega(U) = \min\left(\bar{y}_\omega(a^*) + q_\omega \ln\frac{U}{U_{\text{law}}^{\max}},\ \ln \omega_{\max}\right)$$

U	$U_{\text{law}}^{\max}$	$q_P,\ q_\omega$	P_{rated}	$\omega_{\max}$
Wind speed (m/s)	8.5 m/s band top	Fitted (theory 3 and 1)	2050 kW	17.1 rpm

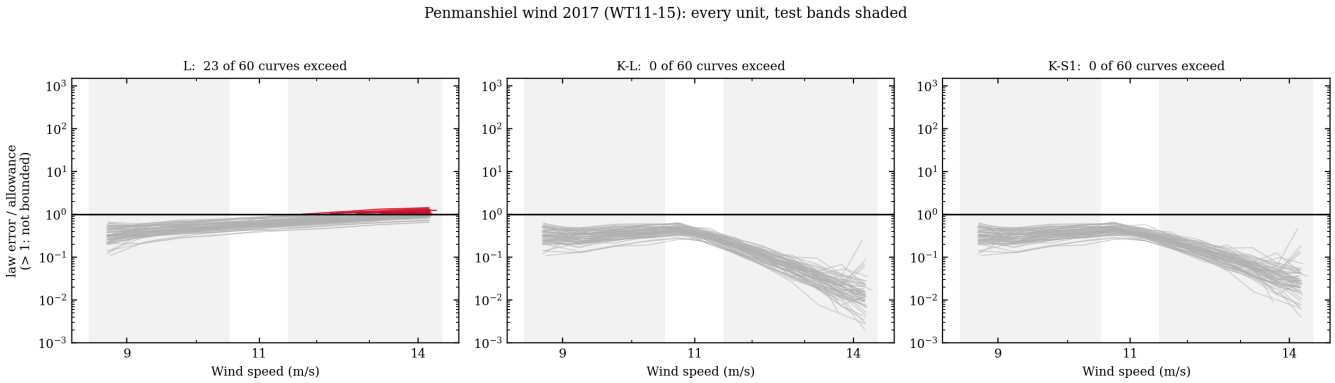
Power and rotor speed follow the law until their published limits (Senvion MM82).

Test band	Rule	Range right	Bound valid	Error proven
below limit	L	100%	100%	0.70
below limit	K-L	100%	100%	0.69
below limit	K-S1	100%	100%	0.69
beyond limit	L	82%	94%	0.78
beyond limit	K-L	100%	100%	0.38
beyond limit	K-S1	100%	100%	0.62

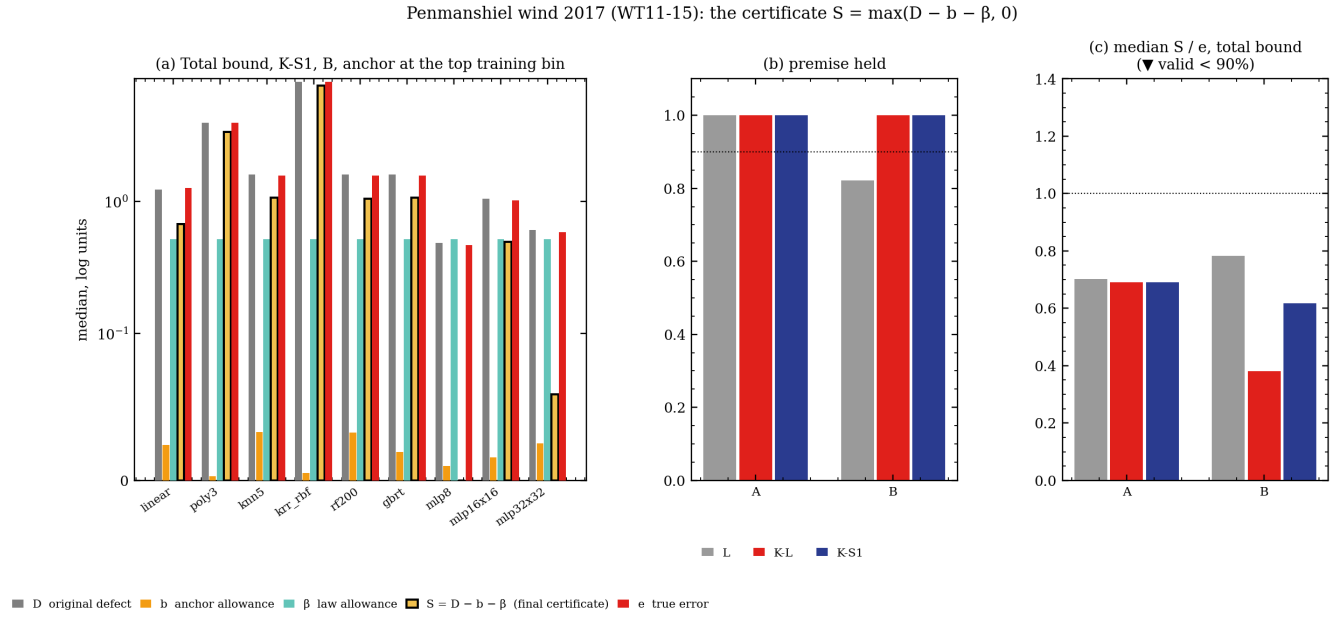
- **82% → 100% range right.**
- **Proves 62% of the error.**



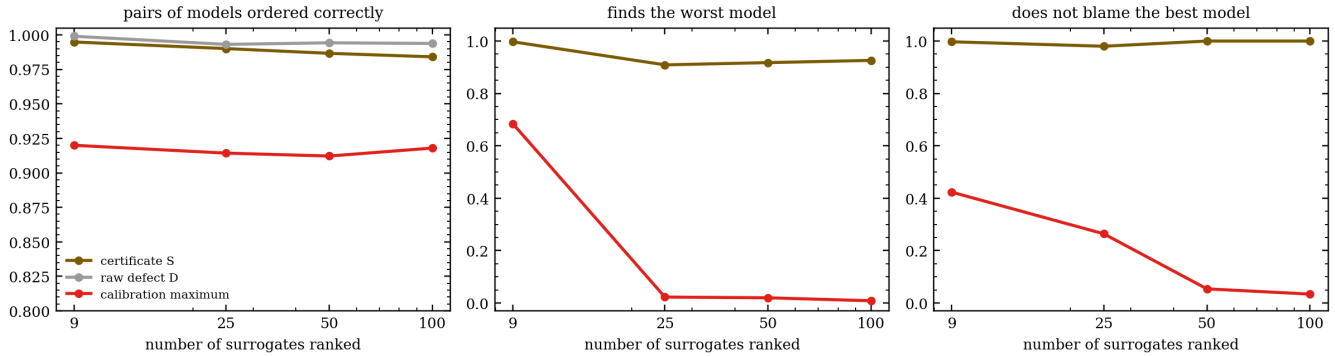
Gold band = allowed range. Red × = point outside it.



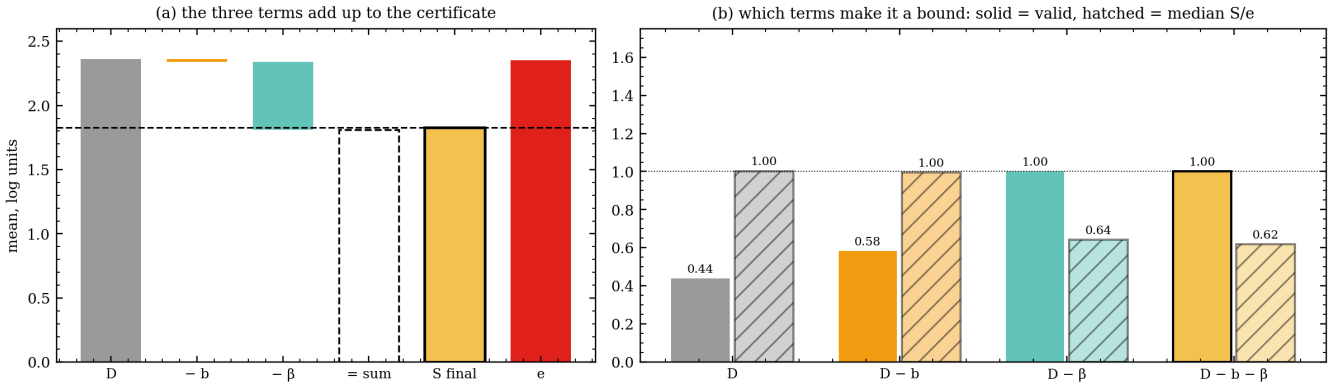
Law error ÷ allowance for every unit. Above 1 = range wrong.



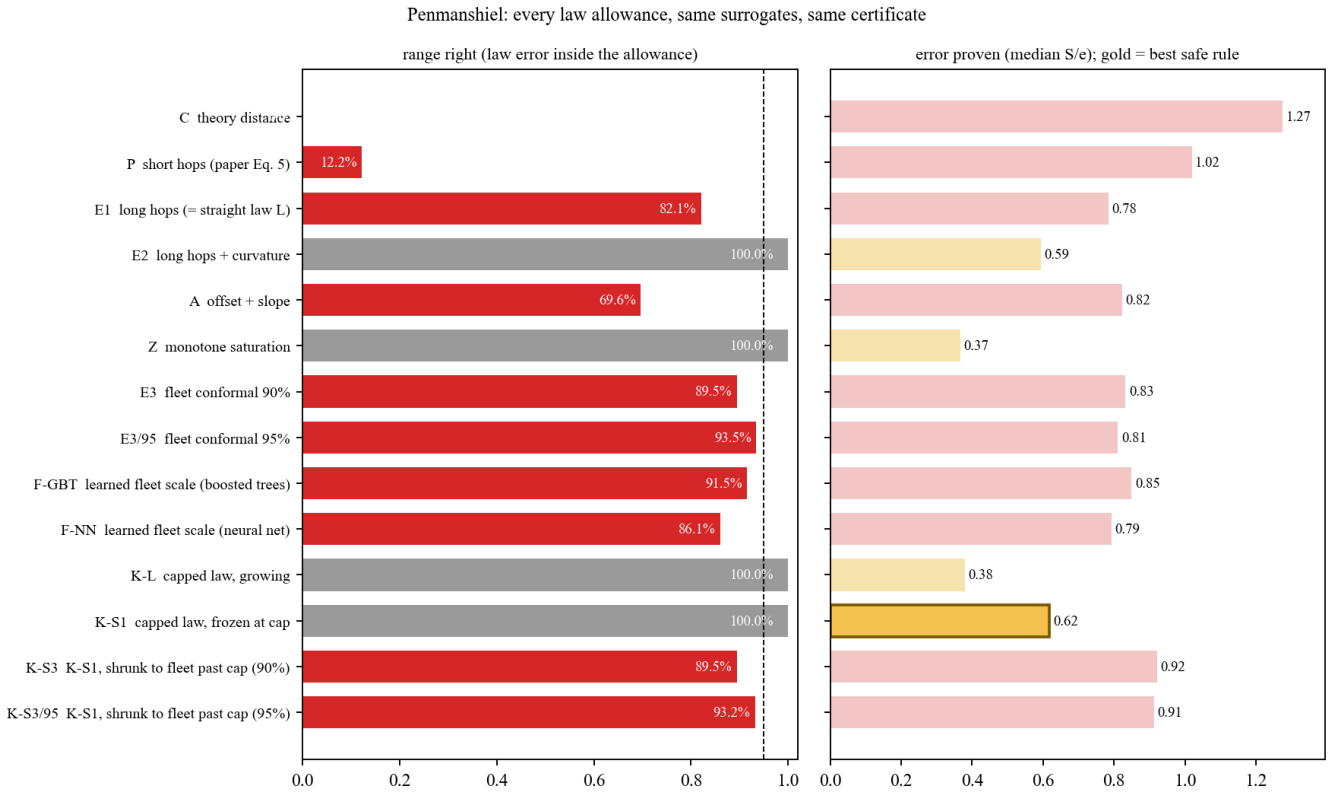
Gold = the certificate S. Red = true error.



Ranking 9 → 100 models: certificate (gold) vs simple baseline (red).



Ablation: D − b − β adds up to the certificate (gold).



All 13 law allowances on this dataset (section 11). Red = unsafe; gold = best safe rule.

5. Hill of Towie wind 2019

New. 21 Siemens SWT-2.3-VS-82 turbines (Scotland). 252 turbine-months. Power only.

Law for this dataset

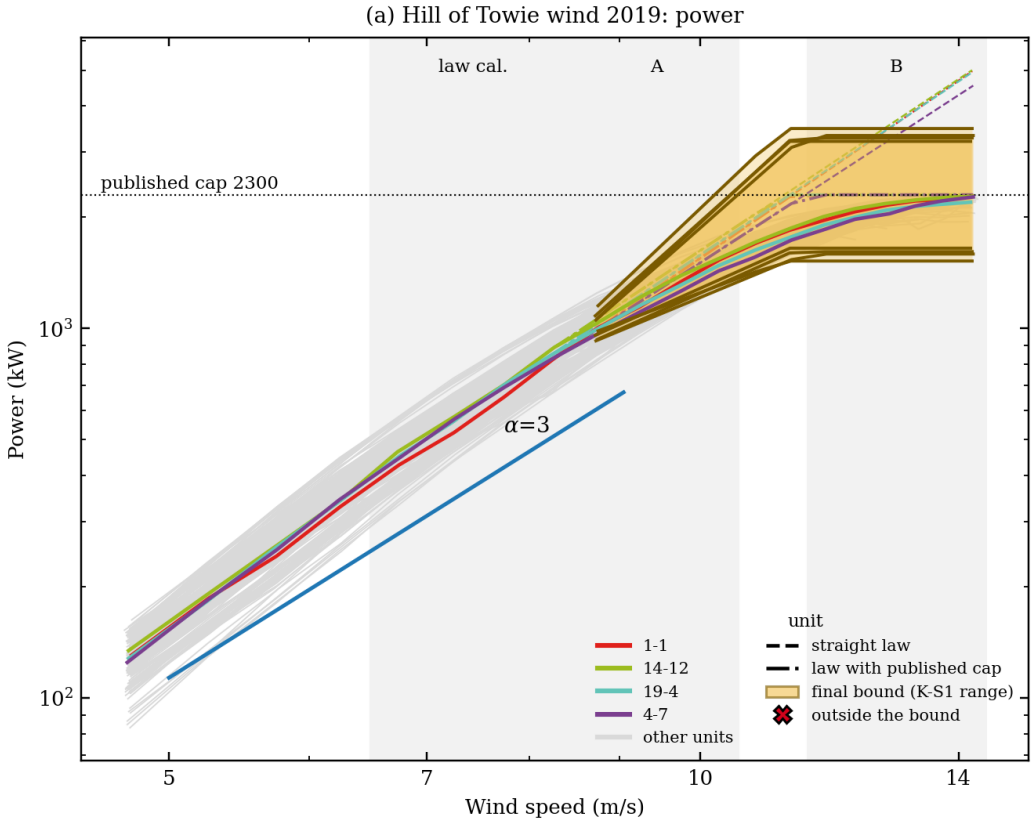
$$c_P(U) = \min\left(\bar{y}_P(a^*) + q_P \ln \frac{U}{U_{\text{law}}^{\max}},\ \ln 2300\ \text{kW}\right)$$

U	$U_{\text{law}}^{\max}$	q_P, q_ω	P_{rated}
Wind speed (m/s)	8.5 m/s band top	Fitted (theory 3 and 1)	2300 kW

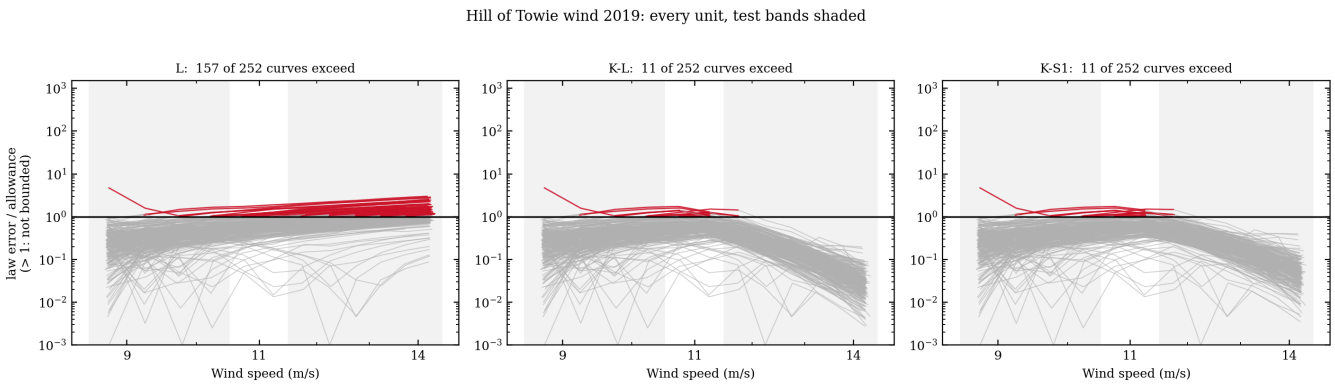
Power follows the cube law until rated power (Siemens SWT-2.3-VS-82; power only).

Test band	Rule	Range right	Bound valid	Error proven
below limit	L	98%	99%	0.78
below limit	K-L	98%	99%	0.78
below limit	K-S1	98%	99%	0.78
beyond limit	L	65%	80%	0.88
beyond limit	K-L	100%	100%	0.60
beyond limit	K-S1	100%	100%	0.72

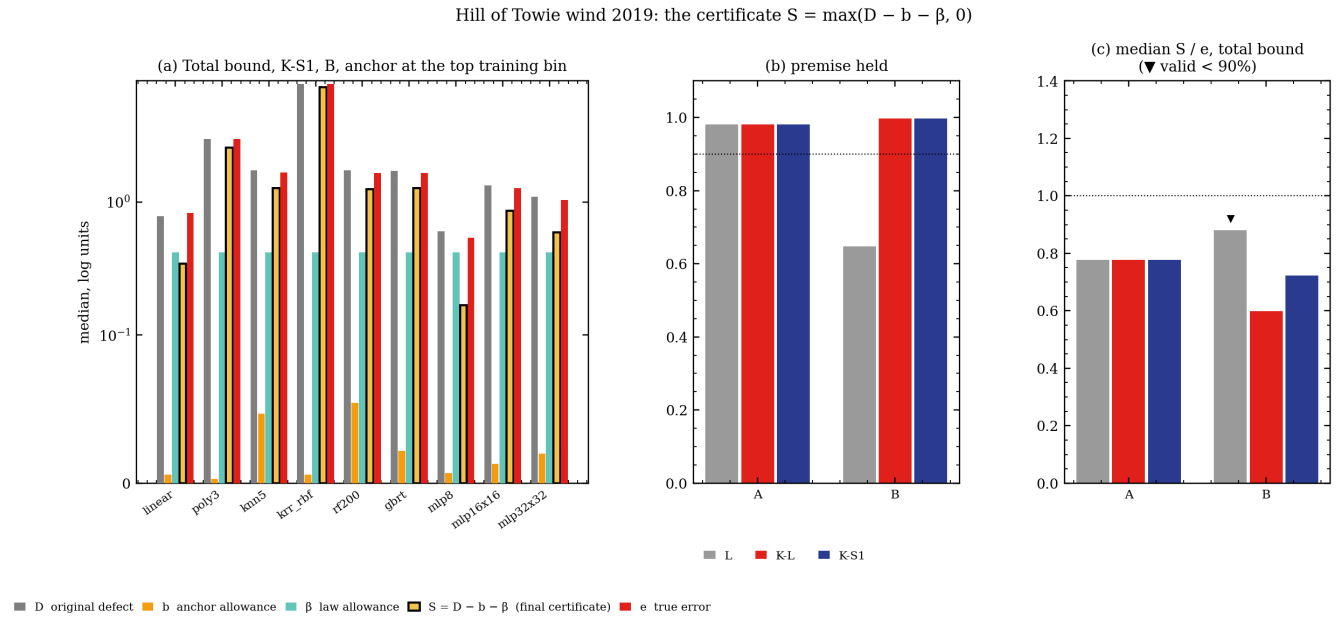
- **65% → 99.7% range right.**
- **Proves 72% of the error.**



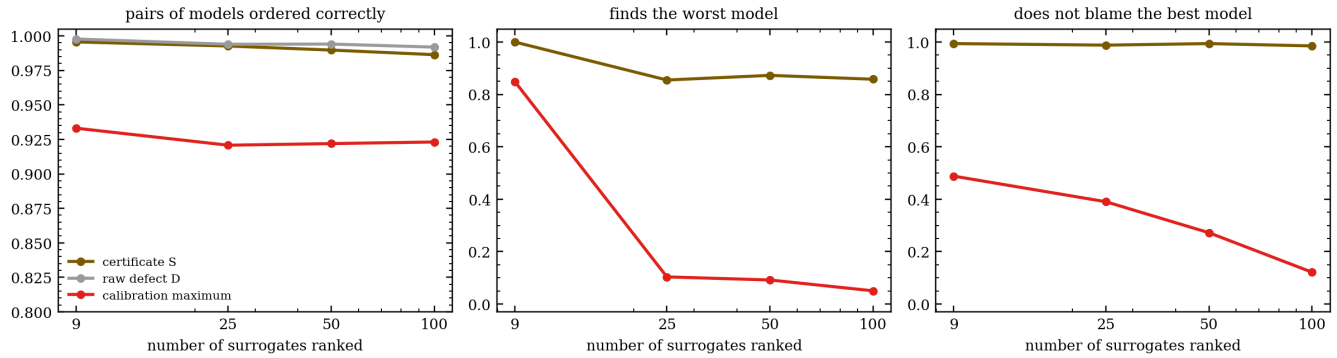
Gold band = allowed range. Red × = point outside it.



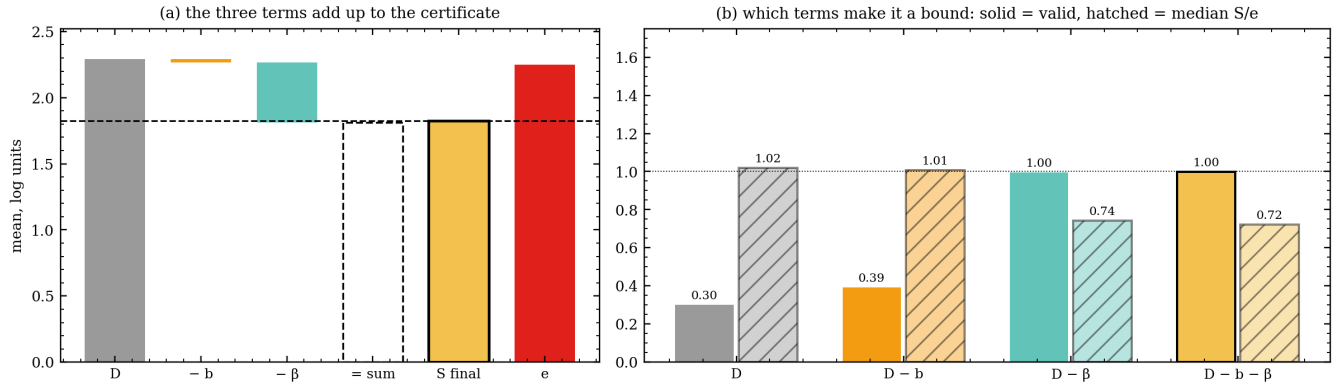
Law error ÷ allowance for every unit. Above 1 = range wrong.



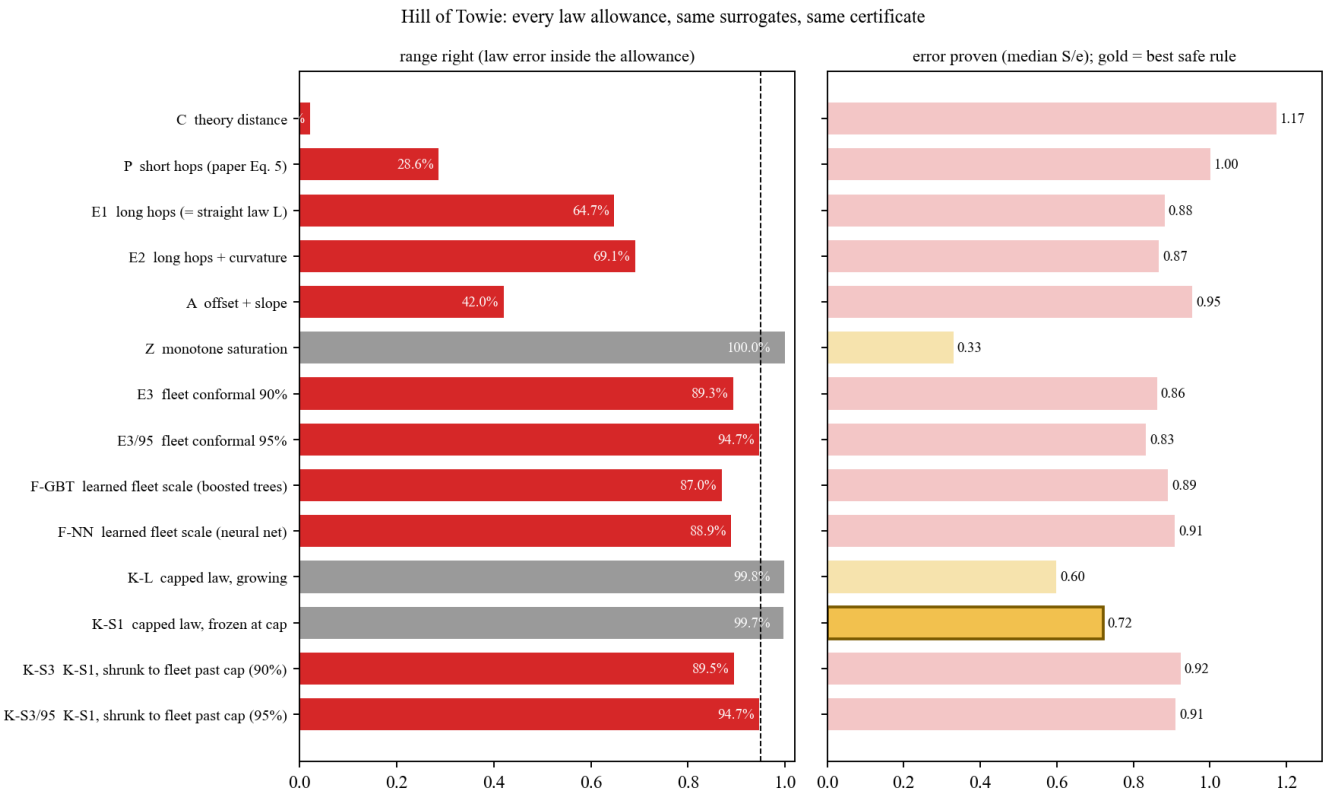
Gold = the certificate S. Red = true error.



Ranking 9 → 100 models: certificate (gold) vs simple baseline (red).



Ablation: D − b − β adds up to the certificate (gold).



All 13 law allowances on this dataset (section 11). Red = unsafe; gold = best safe rule.

6. La Haute Borne wind 2014–15

New. Four Senvion MM82 turbines (France). 93 turbine-months. Power only.

Law for this dataset

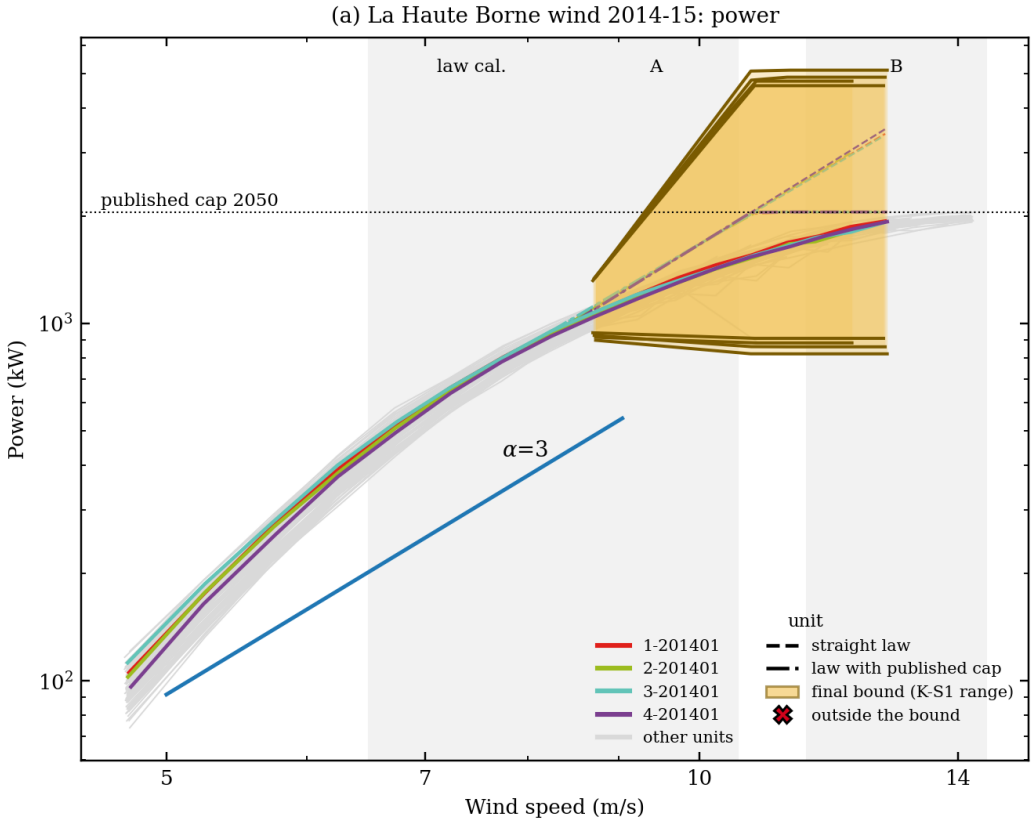
$$c_P(U) = \min\left(\bar{y}_P(a^*) + q_P \ln \frac{U}{U_{\text{law}}^{\max}},\ \ln 2050\ \text{kW}\right)$$

U	$U_{\text{law}}^{\max}$	q_P, q_ω	P_{rated}
Wind speed (m/s)	8.5 m/s band top	Fitted (theory 3 and 1)	2050 kW

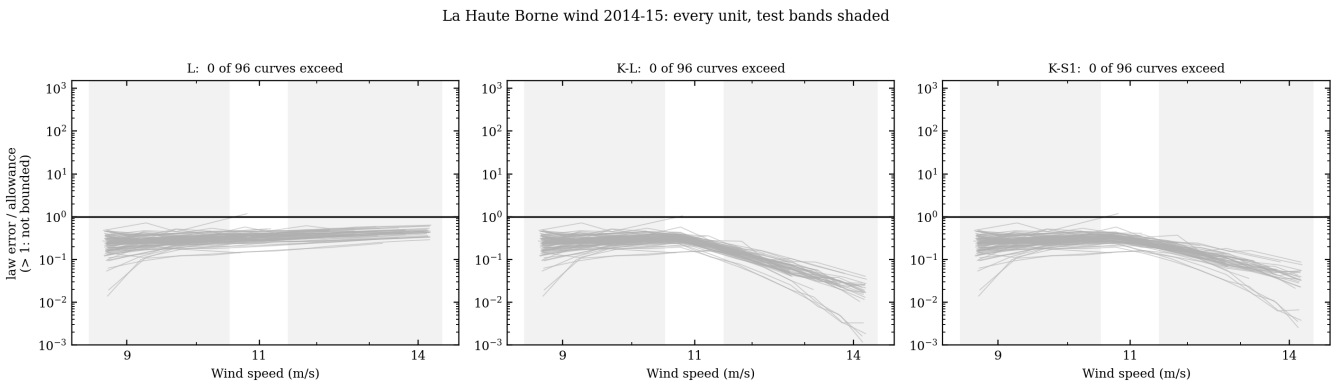
Power follows the cube law until rated power (Senvion MM82; power only).

Test band	Rule	Range right	Bound valid	Error proven
below limit	L	100%	100%	0.41
below limit	K-L	100%	100%	0.41
below limit	K-S1	100%	100%	0.41
beyond limit	L	100%	100%	0.12
beyond limit	K-L	100%	100%	0.00
beyond limit	K-S1	100%	100%	0.28

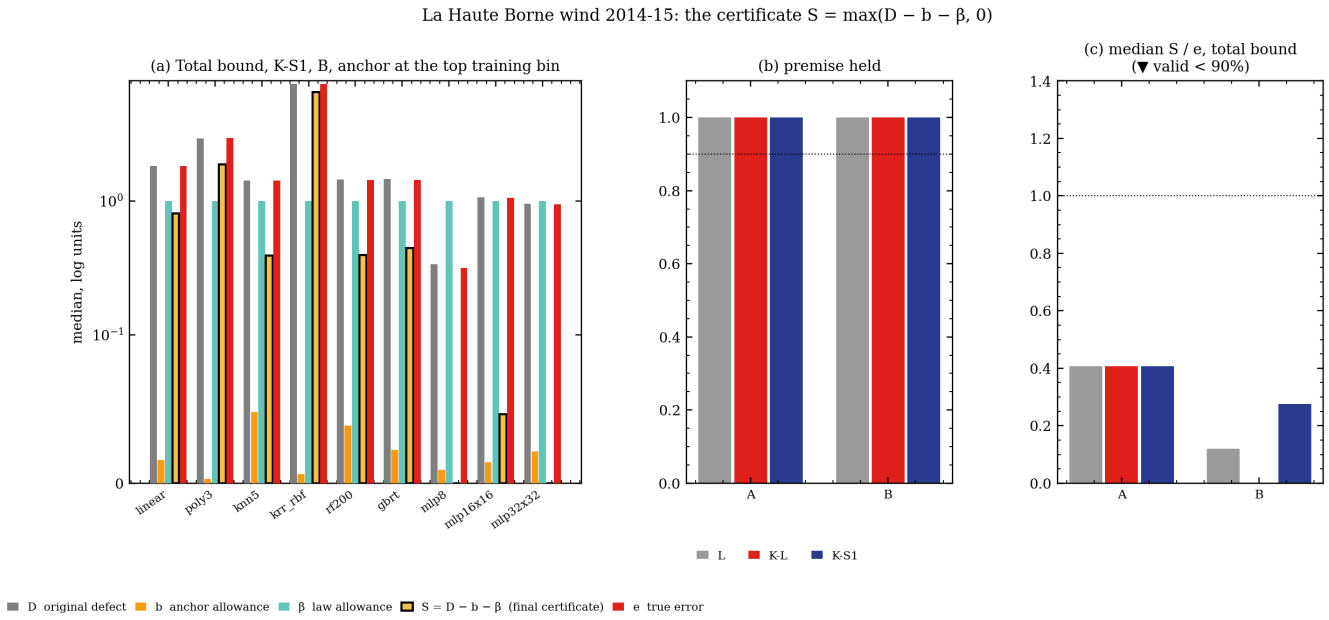
- **Range right everywhere, every rule.**
- **Loose: proves 28% (band ≈ 11× the real law error).**



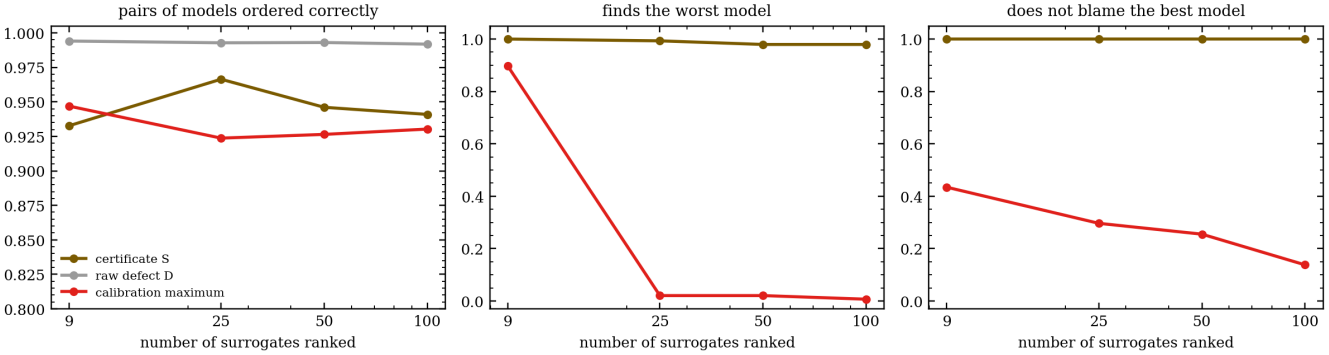
Gold band = allowed range. Red × = point outside it.



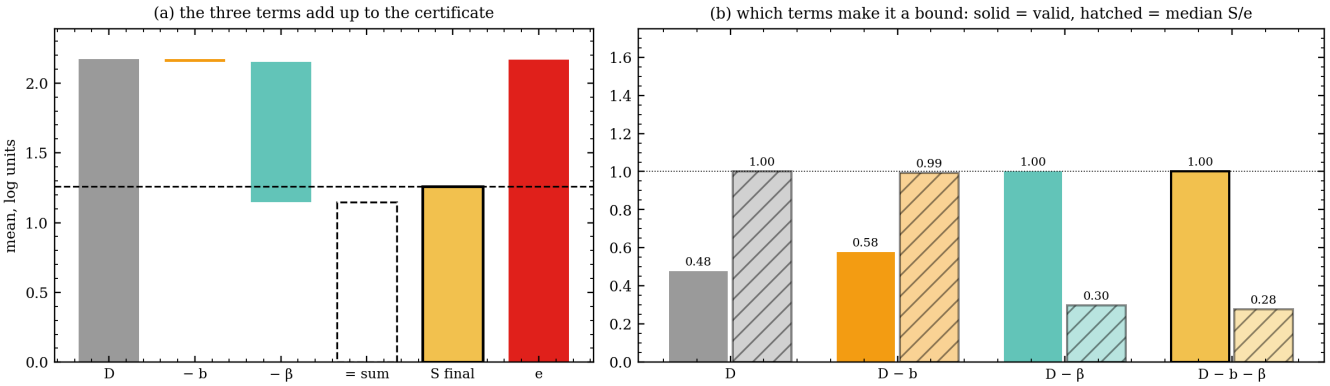
Law error ÷ allowance for every unit. Above 1 = range wrong.



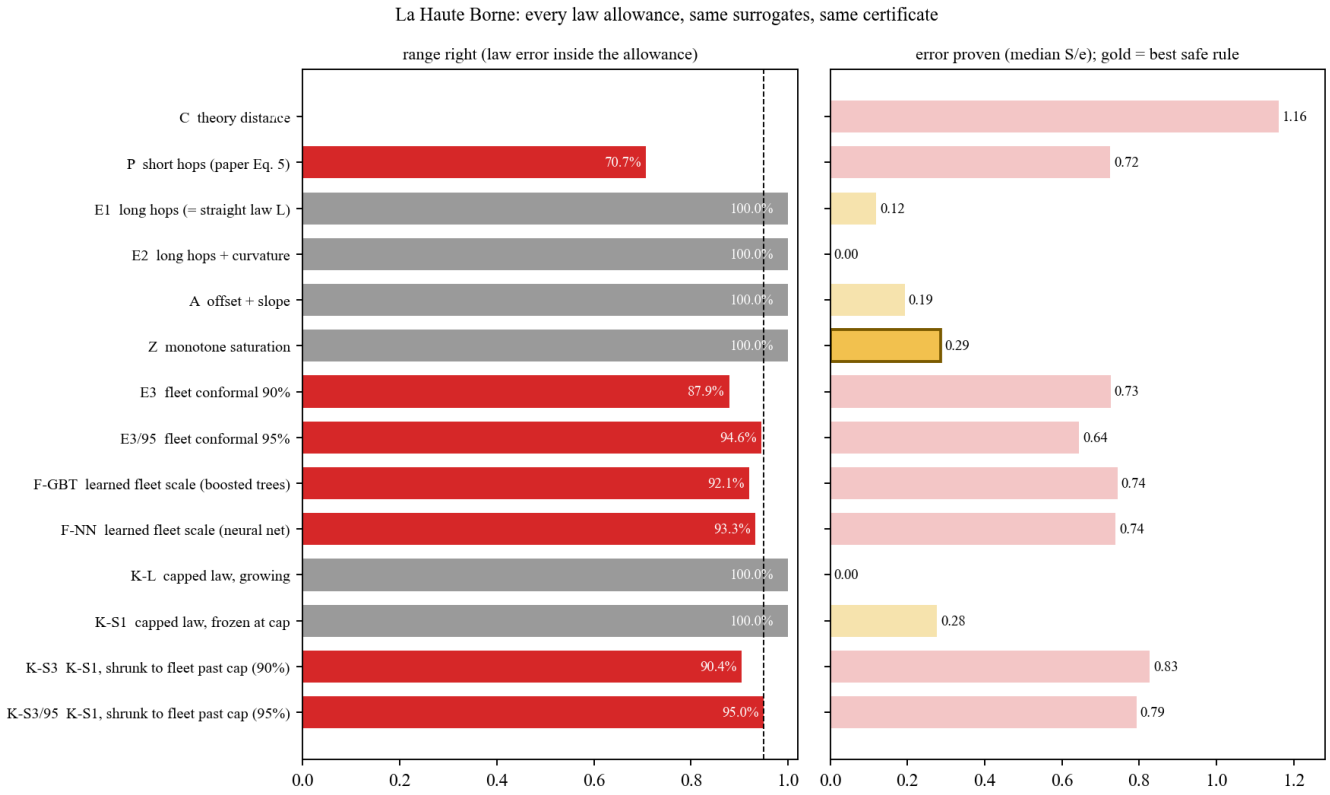
Gold = the certificate S. Red = true error.



Ranking 9 → 100 models: certificate (gold) vs simple baseline (red).



Ablation: D − b − β adds up to the certificate (gold).



All 13 law allowances on this dataset (section 11). Red = unsafe; gold = best safe rule.

7. Solar 2107 (AC power)

New. 893 kW array, 24 ABB TRIO-27.6 inverters (California). 1,671 inverter-months.

Law for this dataset

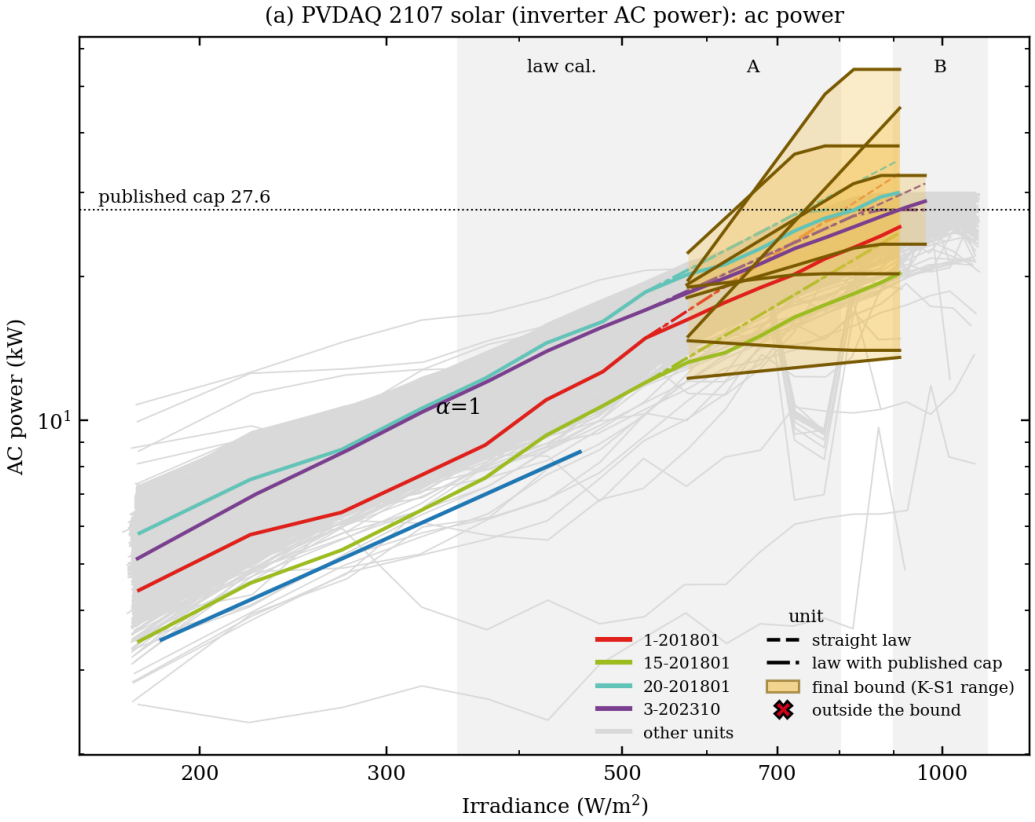
$$c(G) = \min\left(\bar{y}(a^*) + q \ln \frac{G}{G_{\text{law}}^{\max}},\ \ln 27.6\ \text{kW}\right)$$

G	$G_{\text{law}}^{\max}$	q	27.6 kW
Irradiance (W/m²)	550 W/m² band top	Fitted (theory 1)	ABB TRIO-27.6 rating

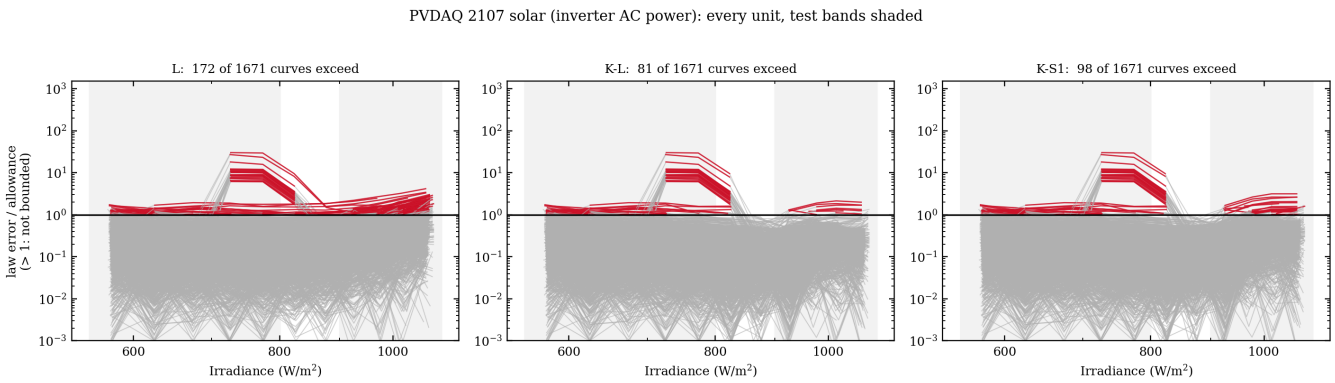
AC power follows irradiance until the inverter reaches its rated 27.6 kW (30 kW checked too).

Test band	Rule	Range right	Bound valid	Error proven
below limit	L	98%	99%	0.61
below limit	K-L	98%	99%	0.60
below limit	K-S1	98%	99%	0.61
beyond limit	L	95%	97%	0.58
beyond limit	K-L	99%	100%	0.43
beyond limit	K-S1	98%	100%	0.53

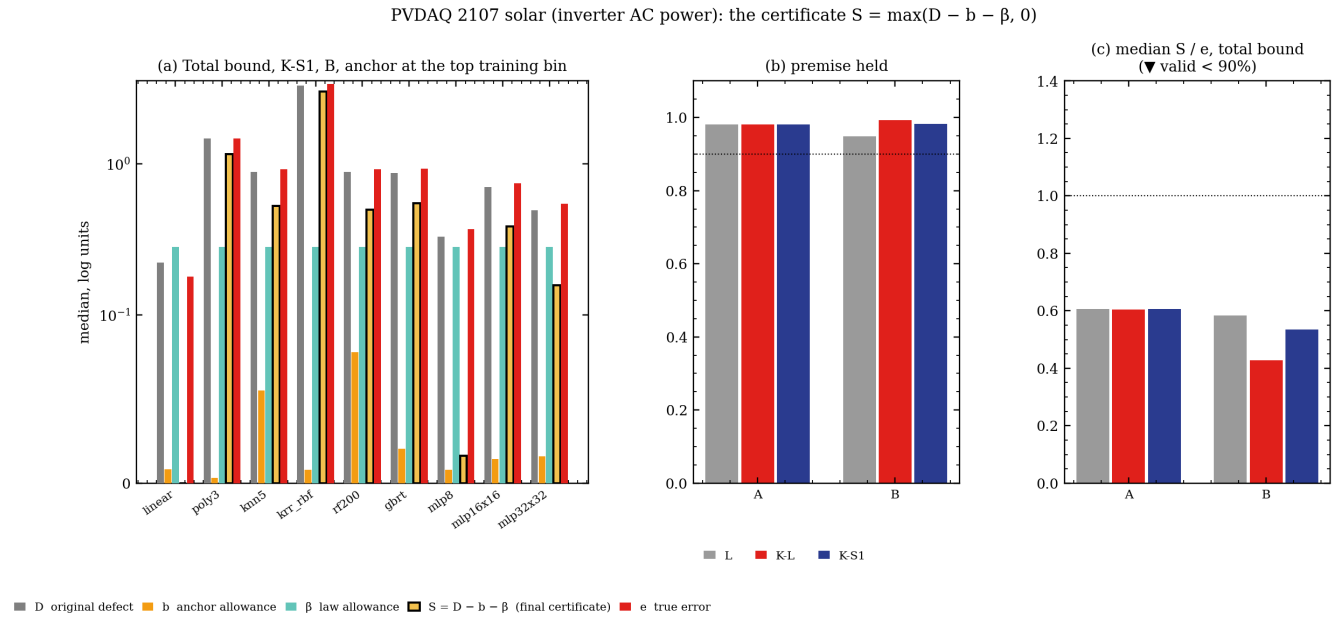
- **98% range right; the inverters rarely reach 27.6 kW.**
- **Proves 53% of the error.**



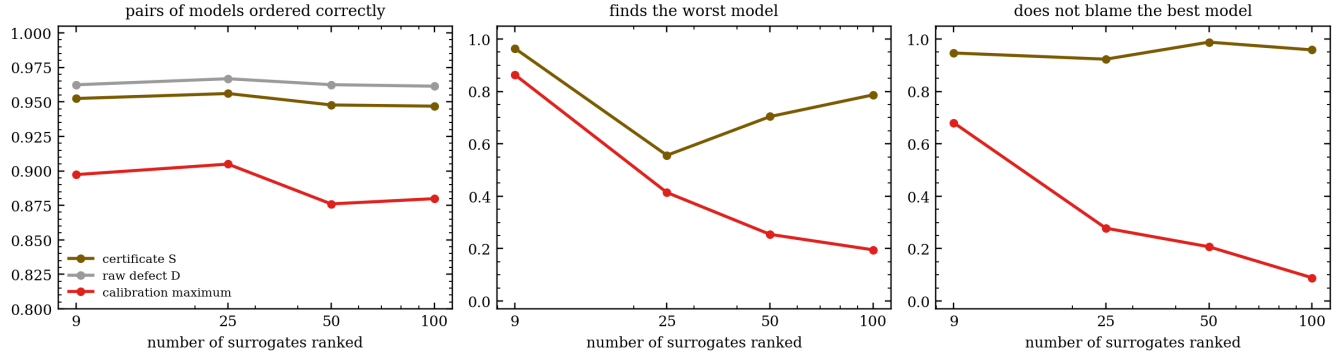
Gold band = allowed range. Red × = point outside it.



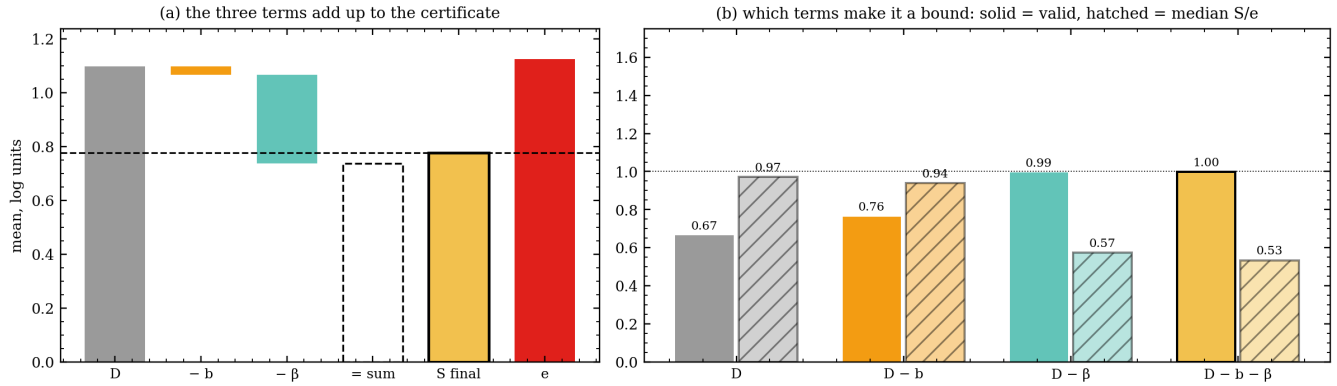
Law error ÷ allowance for every unit. Above 1 = range wrong.



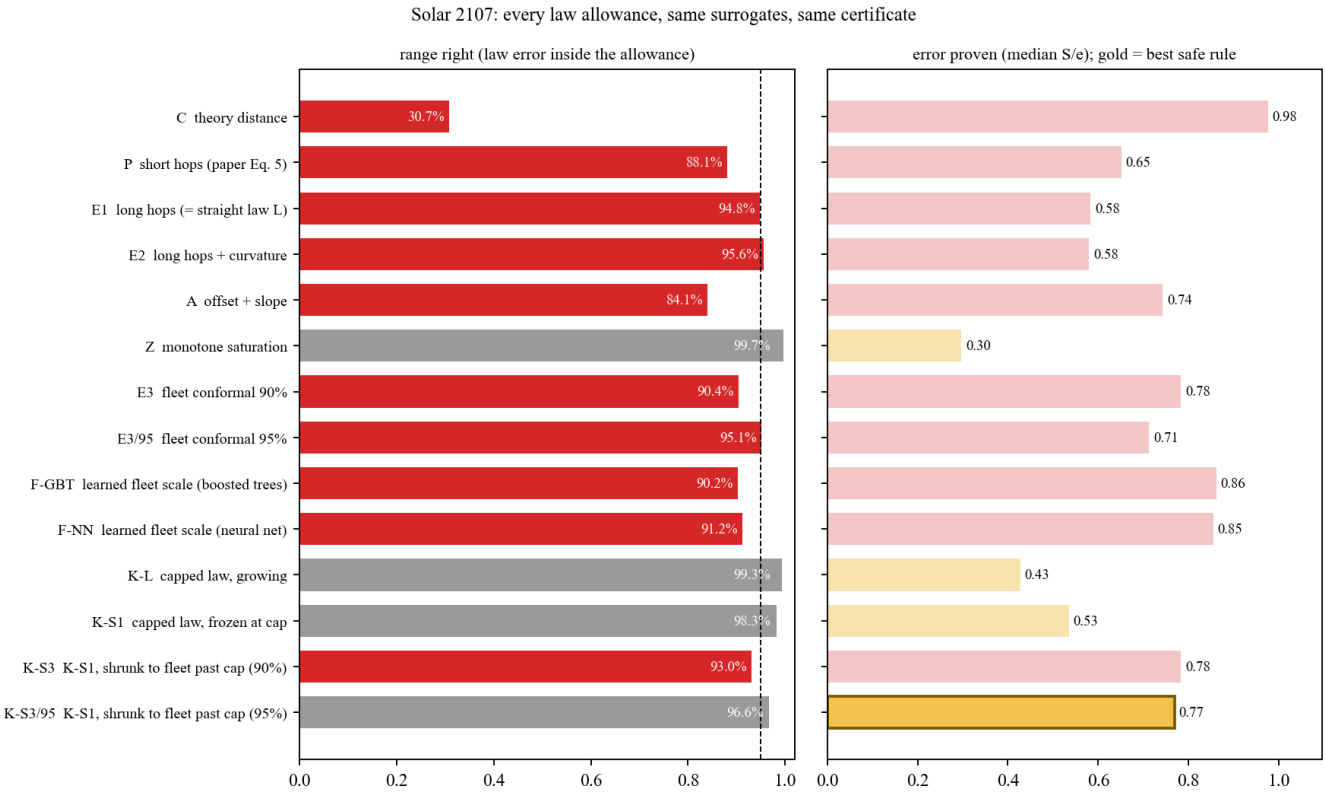
Gold = the certificate S. Red = true error.



Ranking 9 → 100 models: certificate (gold) vs simple baseline (red).



Ablation: D − b − β adds up to the certificate (gold).



All 13 law allowances on this dataset (section 11). Red = unsafe; gold = best safe rule.

8. Solar 9068 (DC current)

4.7 MW tracker plant (Colorado). 384 combiner-months. The limit is on sunlight, not current.

Law for this dataset

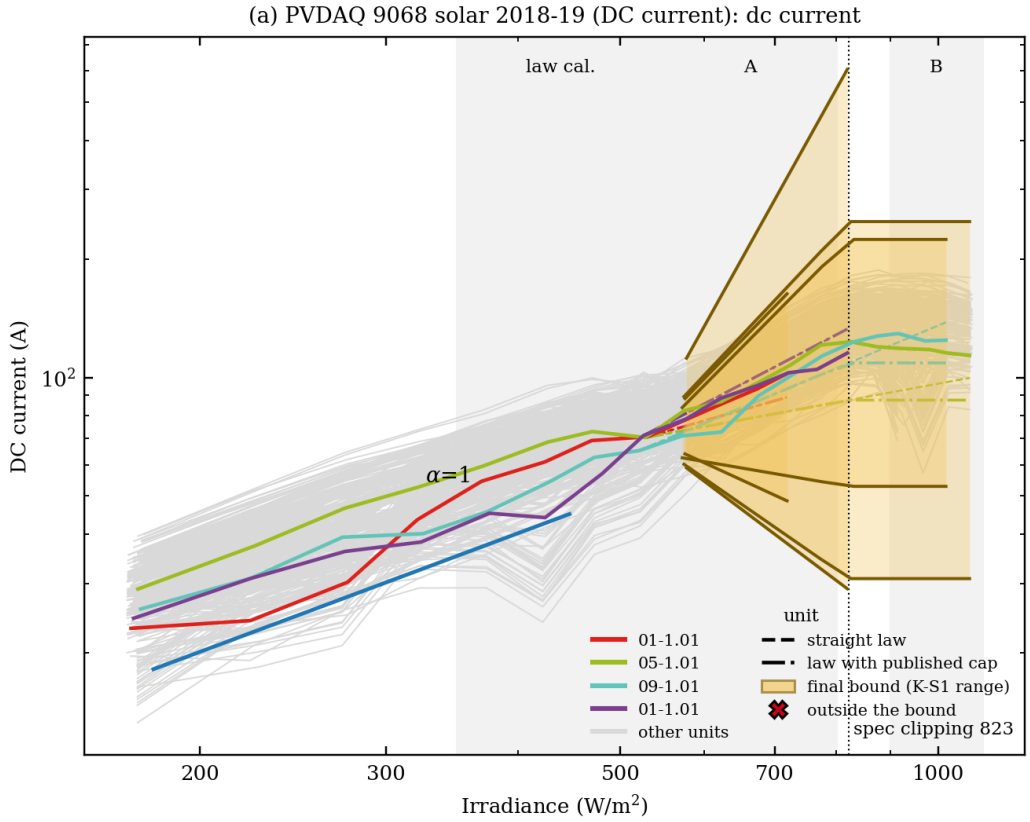
$$c(G) = \bar{y}(a^{*}) + q(\min(\ln G,\ \ln 823) - \ln G_{\text{law}}^{\max})$$

G	$G_{\text{law}}^{\max}$	q	823
Irradiance (W/m²)	550 W/m² band top	Fitted (theory 1)	Inverter limit in irradiance

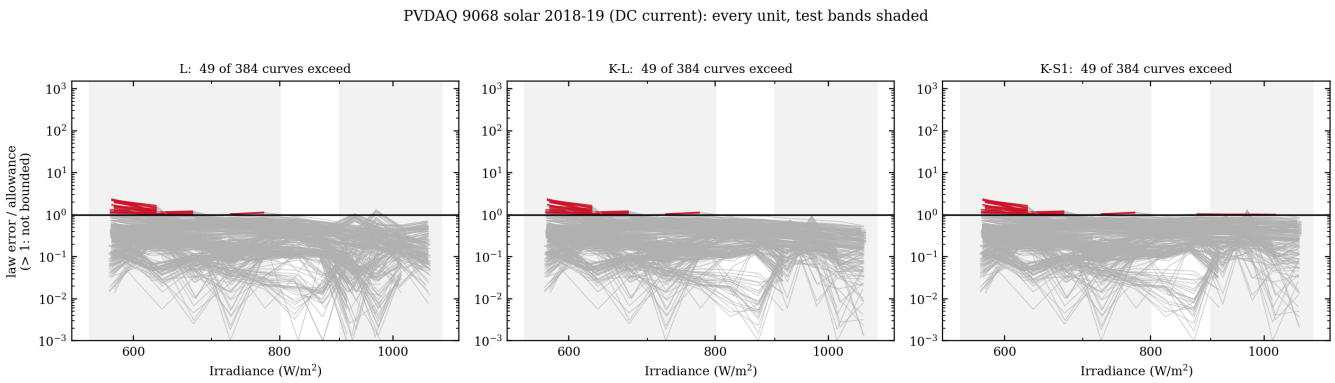
The published limit caps irradiance, not current: 1000 × 1949 / 2368.8 = 823 W/m².

Test band	Rule	Range right	Bound valid	Error proven
below limit	L	95%	98%	0.02
below limit	K-L	95%	98%	0.02
below limit	K-S1	95%	98%	0.02
beyond limit	L	99%	99%	0.00
beyond limit	K-L	100%	100%	0.00
beyond limit	K-S1	99%	100%	0.02

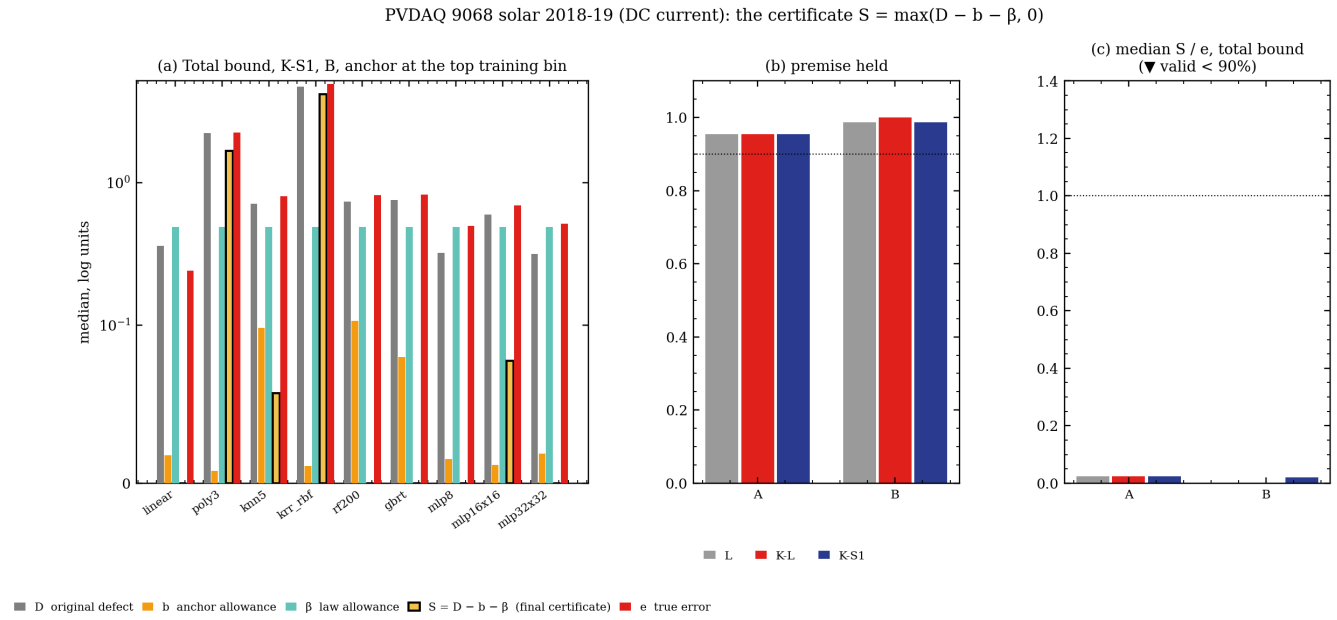
- **Valid, but almost silent: proves 2%.**
- **Needs an output with its own limit (e.g. AC power).**



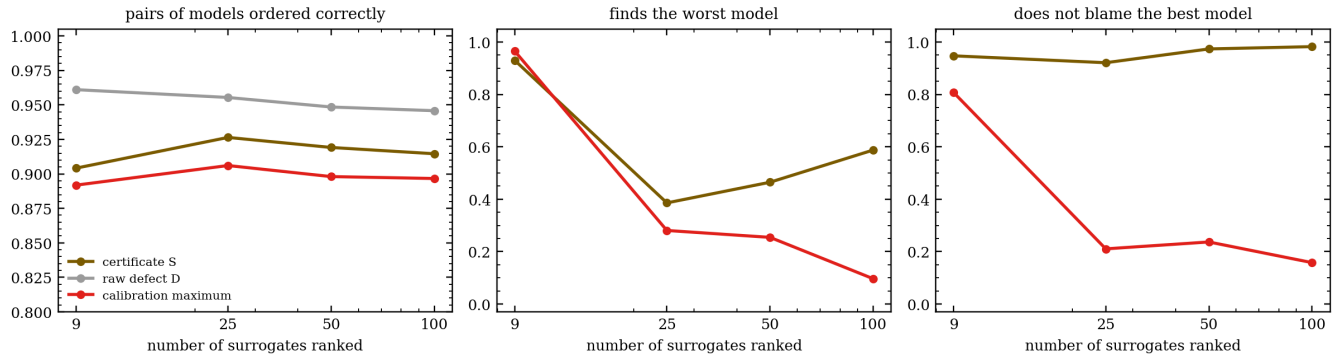
Gold band = allowed range. Red × = point outside it.



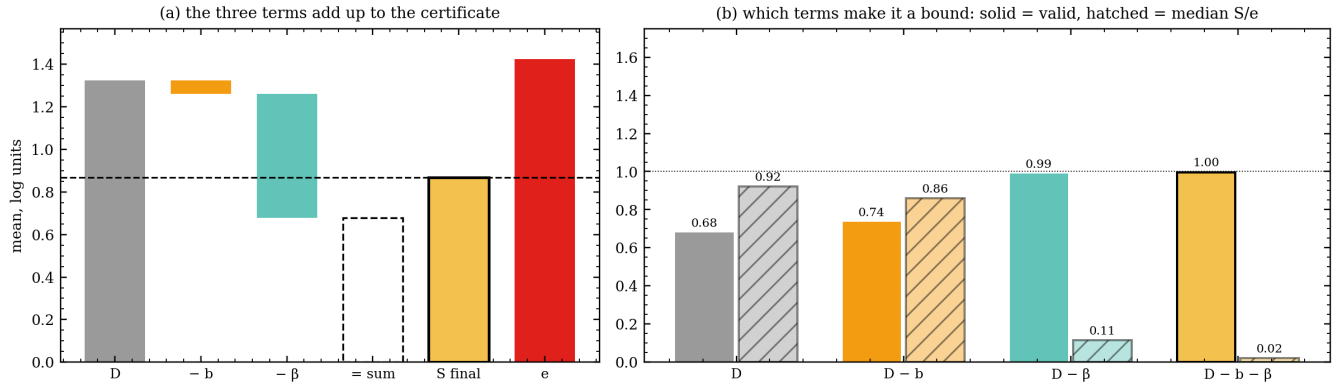
Law error ÷ allowance for every unit. Above 1 = range wrong.



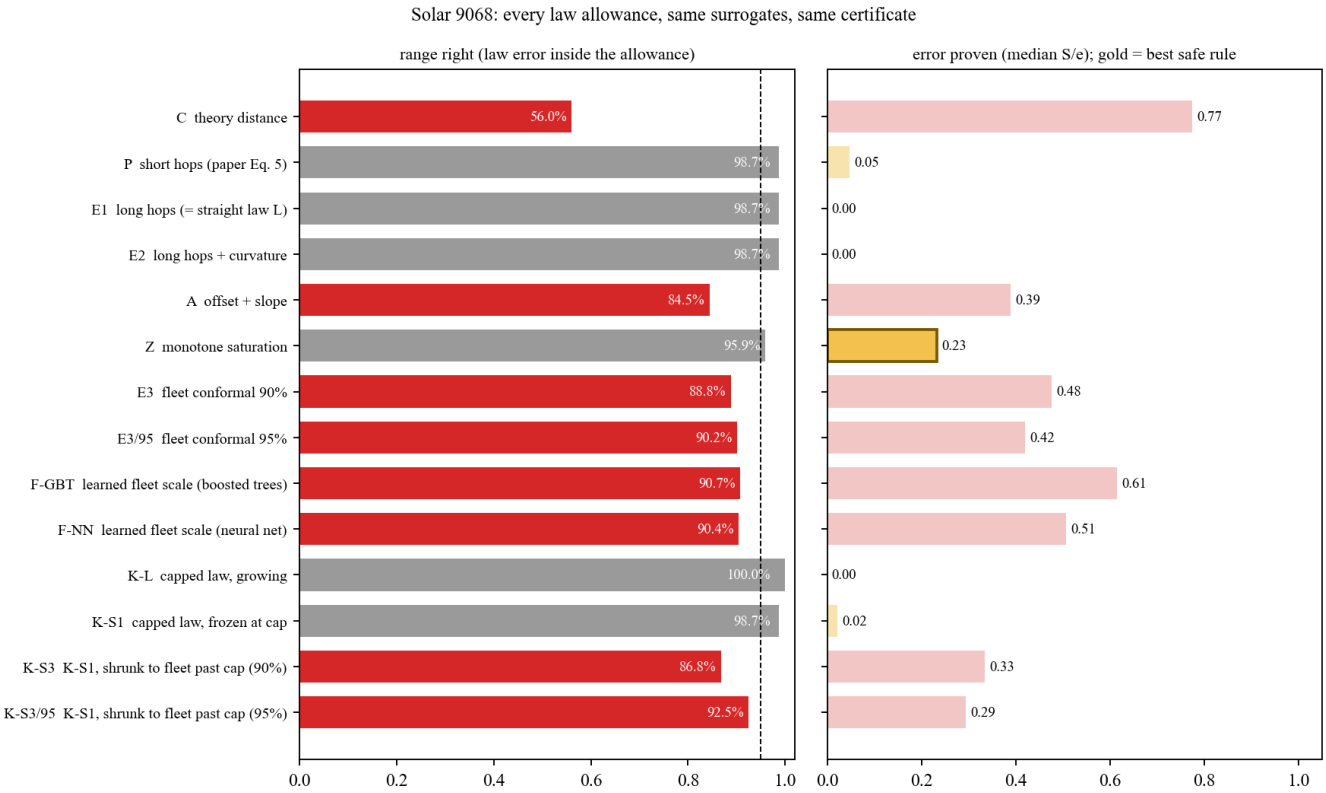
Gold = the certificate S. Red = true error.



Ranking 9 → 100 models: certificate (gold) vs simple baseline (red).



Ablation: D − b − β adds up to the certificate (gold).



All 13 law allowances on this dataset (section 11). Red = unsafe; gold = best safe rule.

9. Capillary waves (lab)

20 lab trials, 7–350 mW. 12-band spectra. No published limit, so K-S1 = straight law.

Law for this dataset

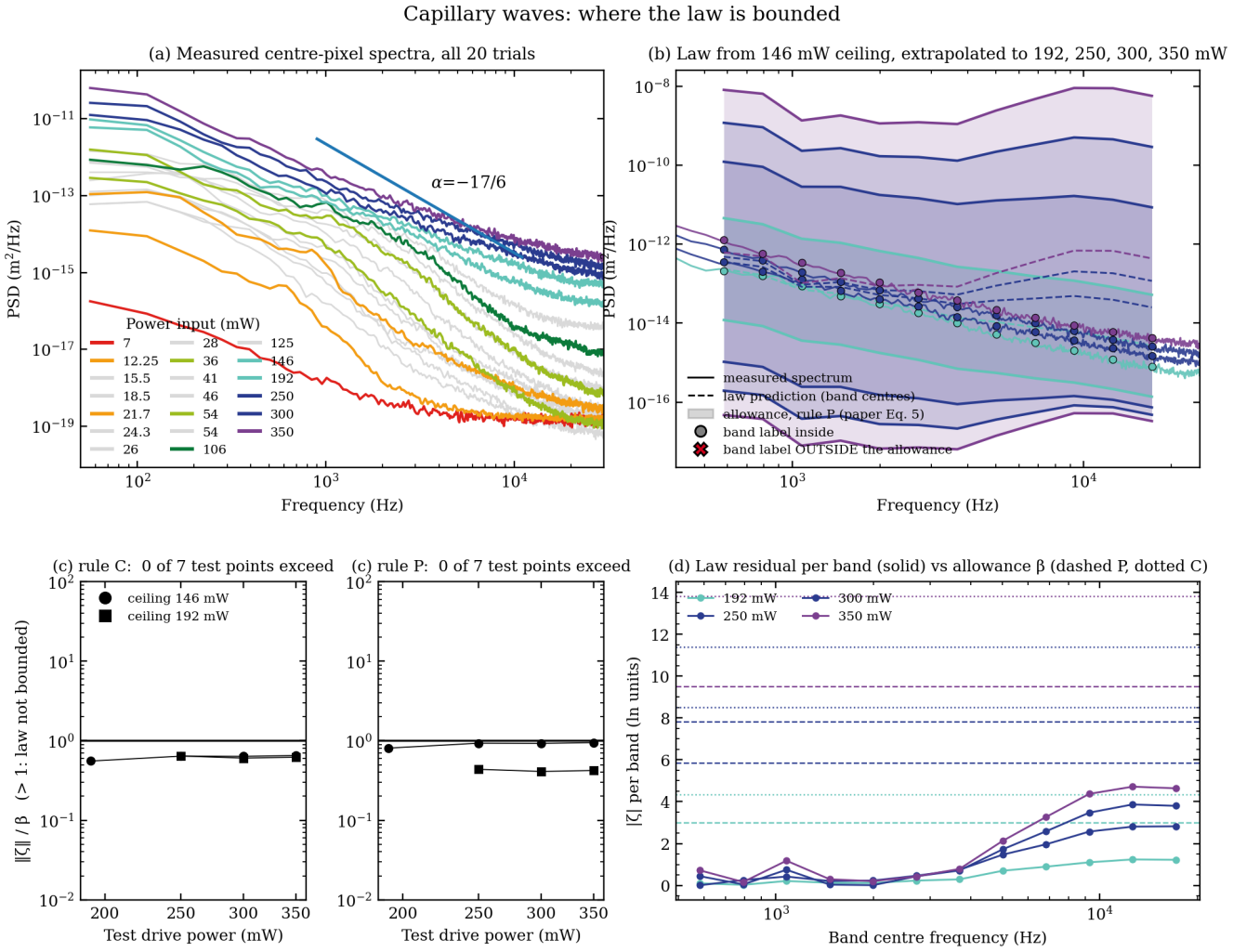
$$c_j(P) = \bar{y}_j(a^*) + q_j \ln \frac{P}{P_{\text{law}}^{\max}}, \quad j = 1, \ldots, 12$$

P	$P_{\text{law}}^{\max}$	$\bar{y}_j$	q_j	y_{cap}
Drive power (mW)	146 or 192 mW	ln PSD in band j	Fitted exponent per band	none published

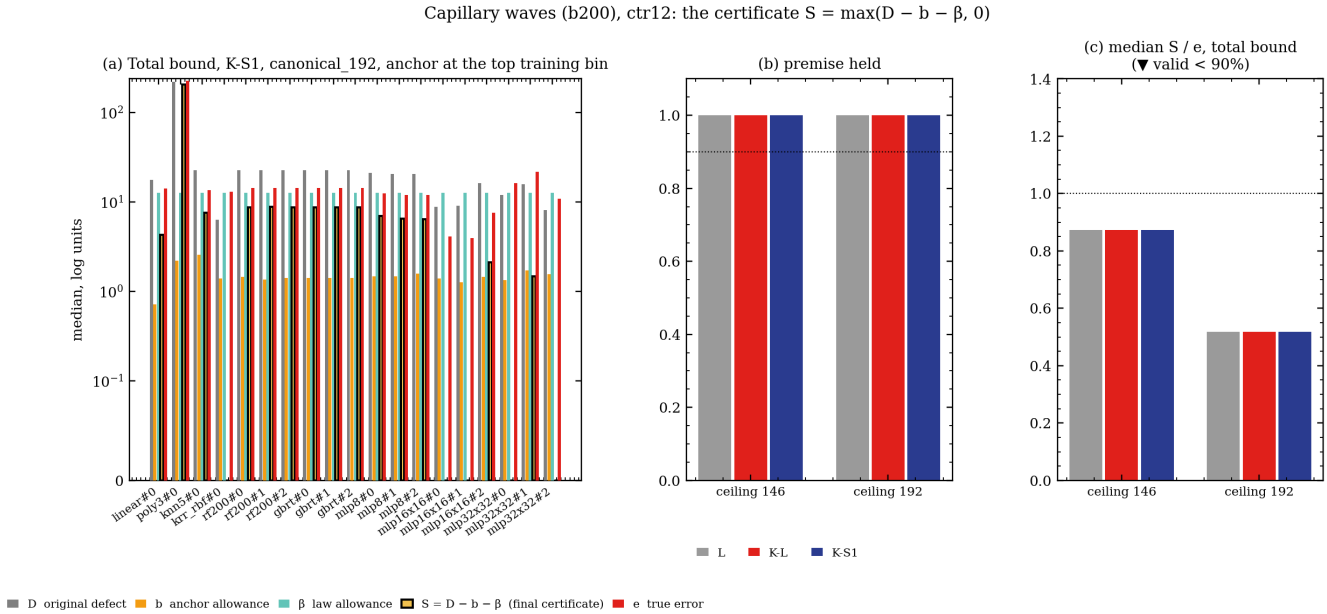
No published limit above the ceiling, so the law is a straight line and nothing freezes.

Test band	Rule	Range right	Bound valid	Error proven
ceiling 146 mW	L = K-S1	100%	100%	0.87
ceiling 192 mW	L = K-S1	100%	100%	0.52

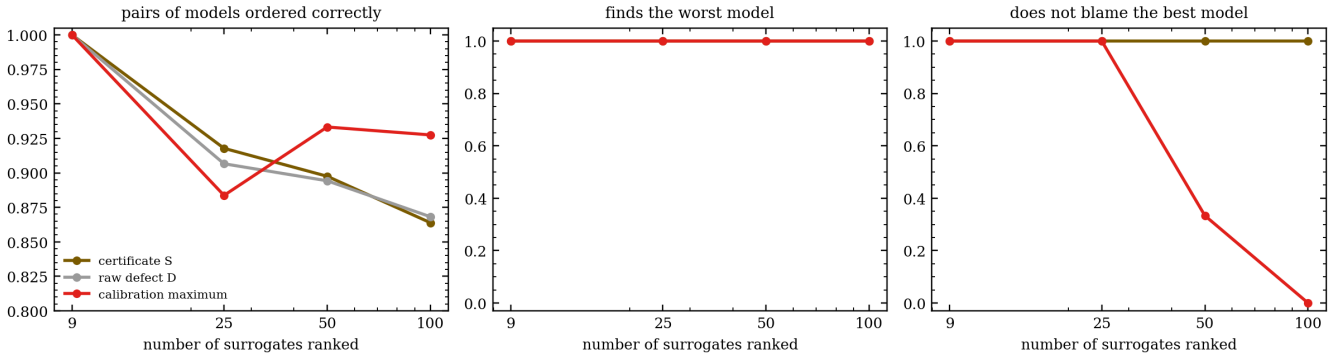
- **Valid on every test trial (only 3–4 trials).**
- **The anchor allowance b is essential here.**



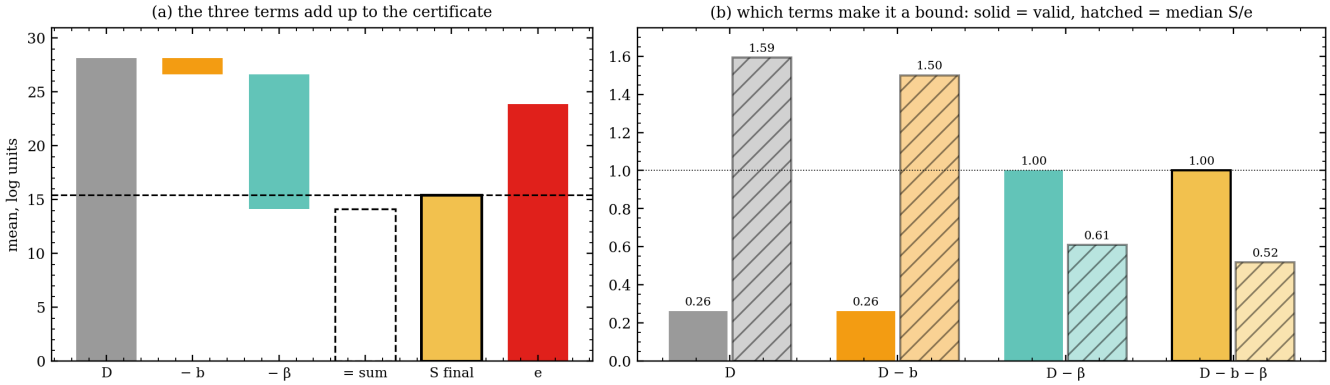
Spectra, the law and its allowance (band) at each test trial.



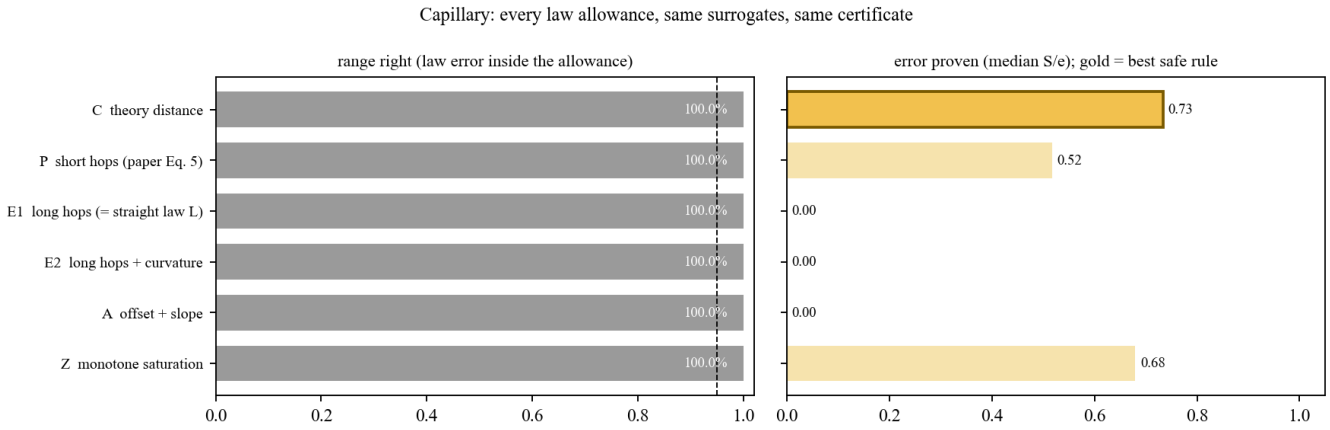
Gold = the certificate S. Red = true error.



Ranking 9 → 100 models: certificate (gold) vs simple baseline (red).



Ablation: D − b − β adds up to the certificate (gold).



All 13 law allowances on this dataset (section 11). Red = unsafe; gold = best safe rule.

10. Tighter beyond the cap: K-S3

K-S1 stays as wide past the cap as at the knee. K-S3 shrinks it to the plateau error seen on the same turbines in earlier years. Designed on Kelmarsh 2017–2020; **confirmed on 2021**, untouched until the rule was fixed.

K-S3: shrink the allowance after the cap

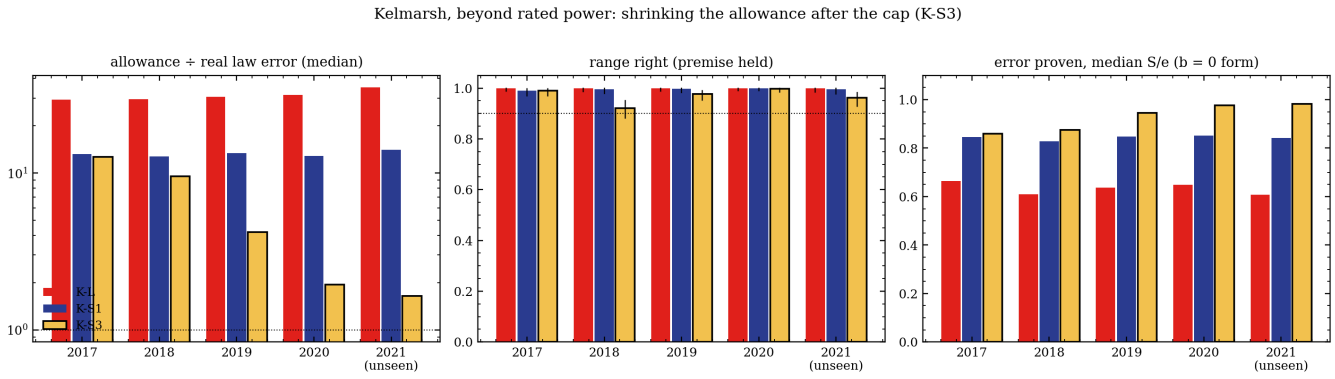
$$\beta_\lambda(x) = \kappa'\delta \;\; (\delta \le \delta^*), \qquad \beta_\lambda(x) = \min\left(\kappa'\delta^*,\; q_{90}^{\text{fleet}}\right) \;\; (\delta > \delta^*)$$

$\kappa'\delta$	δ^*	q_{90}^{fleet}	min
K-S1 below the cap	where the law hits its cap	90% conformal quantile, earlier years	never wider than K-S1

Past the cap the allowance shrinks to the plateau error seen on the same turbines in earlier years.

Rule (2021, beyond rated)	Allowance ÷ real error	Range right [95% CI]	Bound valid	Error proven	Error proven (b = 0)
K-L	35.1×	100% [98%–100%]	100%	0.54	0.61
K-S1	14.1×	100% [98%–100%]	100%	0.77	0.84
K-S3	1.6×	96% [93%–98%]	99%	0.93	0.98

- **Allowance 14× → 1.6× the real law error.**
- **Error proven 0.77 → 0.93; still valid on 99% of rows.**
- Needs a history of the same machines; with only 2 earlier years (2018) coverage was 92%.



Beyond rated power, 2017–2021. Gold = K-S3.

11. Every law allowance on every dataset

Same units, same 9 surrogates, same three-term certificate. Only β changes. Fleet rules learn from **other machines in other months**, never the scored machine.

> **Safe** = range right on ≥ 95% of unit-bins **and** bound valid on ≥ 99% of rows.
>
> **Best** = highest error proven among safe rules.



Error proven beyond the published limit, every rule × dataset. Boxed = best safe rule. × = unsafe (number = range right).

Dataset	Best safe rule	Range right	Bound valid	Error proven	K-S1 before
Kelmarsh	**K-S3/95**	96%	99%	**0.94**	0.80
Penmanshiel	**K-S1**	100%	100%	**0.62**	0.62
Hill of Towie	**K-S1**	100%	100%	**0.72**	0.72
La Haute Borne	**Z**	100%	100%	**0.29**	0.28
Solar 2107	**K-S3/95**	97%	99%	**0.77**	0.53
Solar 9068	**Z**	96%	100%	**0.23**	0.02
Capillary	**C**	100%	100%	**0.73**	0.52

Capillary has no published limit: last column is the paper rule P.

- **0 bound failures where the range was right**, all 13 rules, all 7 datasets.
- **K-S3 (fleet, 95%) wins where it is safe:** Kelmarsh 0.80 → 0.94, Solar 2107 0.53 → 0.77.
- **K-S3 sits on the 95% edge elsewhere:** 93–95% range right on Penmanshiel, Hill of Towie, La Haute Borne.
- **K-S1 and Z are the guaranteed fallbacks:** 96–100% range right, prove less.
- **Single-unit rules (C, P, E1, A) fail beyond the limit** on wind: often under 30% range right.
- **Learned fleet scales (trees, neural net)** do not beat the plain fleet quantile.
- **Solar 9068 stays loose with every rule** (best 0.23): the law itself is uncertain there.

Exploratory: the best rule was picked after seeing these results. Only K-S3 on Kelmarsh 2021 (section 10) was fixed before its test data were opened. Code: method_compare.py.

12. Real neural surrogates: PDE solvers

The slides' question: check a learned surrogate of a solver **without running the solver**. Fourier Neural Operators [19] (the architecture PDEBench [18] benchmarks), residual CNNs and MLPs were trained on exact solutions of three equations (PDEBench-style initial conditions), then used at parameters **beyond their training range**. The solver is used only afterwards, to check the certificate.

Exact symmetry law (PDE surrogates)

$$u(x,\,t;\,p) = u(x,\,\lambda t;\,p/\lambda) \quad \Rightarrow \quad T_\lambda(u_0, p, t) = (u_0,\; p/\lambda,\; \lambda t),\;\; A_\lambda = I,\;\; c_\lambda = 0$$

p	t	λ	u_0	$p\,t$
speed b or diffusivity D	time	transform parameter	initial condition	invariant (unchanged by T)

Advection and diffusion depend on p and t only through p·t, so an out-of-range p maps to an in-range anchor.

System	Law	Train	Test (beyond training)
Advection $u_t + b\,u_x = 0$	exact	b 0.1–1	b 1.5–4
Diffusion $u_t = D\,u_{xx}$	exact	D 0.002–0.02	D 0.03–0.08
Advection–diffusion (ν = 0.003)	approximate (needs β)	b 0.1–1	b 1.5–4

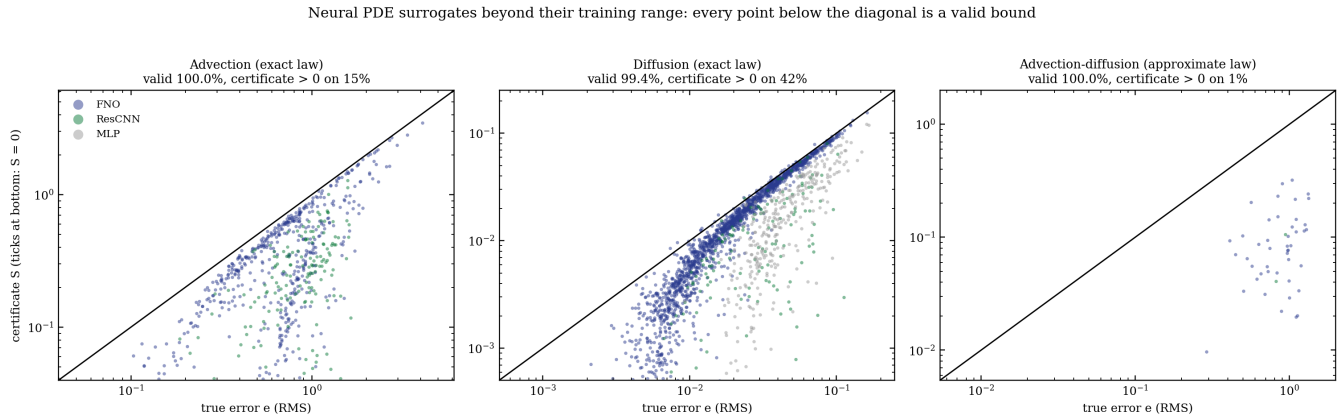
12 surrogates per system (6 FNO, 4 ResCNN, 2 MLP); 1D, 128 points.

System	Valid (global b)	Says som ething	Valid (local b)	Says som ething	Error proven when > 0	Lipschitz says something	Conform al covers
Advection, first test set	100%	8%	100%	14%	0.39	0%	61%
Advection, fresh test set	100%	7%	100%	15%	0.45	0%	63%
Diffusion, first test set	100%	36%	99%	42%	0.65	0%	36%
Diffusion, fresh test set	100%	36%	99%	42%	0.63	0%	36%
Adv.–diffusion, first test set	100%	1%	100%	1%	0.08	0%	33%
Adv.–diffusion, fresh test set	100%	1%	100%	1%	0.09	0%	33%

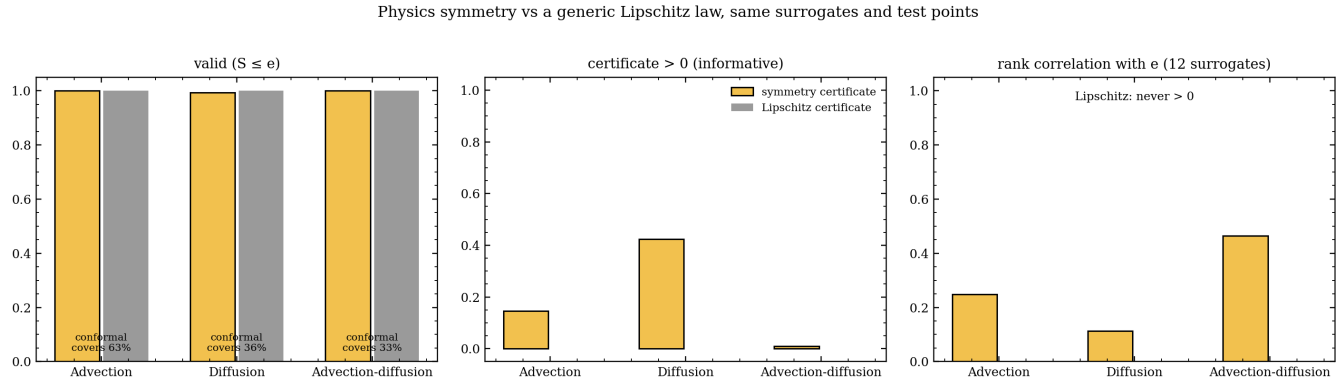
Global b = the predeclared anchor allowance (one 97.5% quantile). Local b = computed within the same p·t range (post hoc rule, then checked on a fresh test set).

- **Never over-claimed: valid on 99.3–100% of points**, every system, both test sets.
- **Diffusion: proves 63–65% of the error when it speaks** (FNOs: median 45–47%).
- **A generic Lipschitz law proves nothing** (0%); the physics symmetry does.

- **In-range conformal bounds fail out of range** (cover 33–63% of errors).
- Advection: rare in-range failures make b large, so it is mostly silent. Advection–diffusion: the law allowance for the approximate law is too wide, so it is silent.



Certificate against true error for every surrogate and test point. Below the diagonal = valid.



Physics symmetry vs a generic Lipschitz law, same surrogates.

13. How sure are these numbers?

Rows from one turbine or inverter are not independent, so whole machines were resampled (2,000 bootstrap draws). The honest sample size is the number of machines.

Dataset (K-S1, beyond limit)	Machines	Range right	Bound valid	Error proven
Kelmarsh	6	100.0% [100.0%–100.0%]	100.0% [100.0%–100.0%]	0.80 [0.78–0.81]
Penmanshiel	5	100.0% [100.0%–100.0%]	100.0% [100.0%–100.0%]	0.62 [0.60–0.64]
Hill of Towie	21	99.7% [99.5%–100.0%]	99.9% [99.8%–100.0%]	0.72 [0.71–0.73]
La Haute Borne	4	100.0% [100.0%–100.0%]	100.0% [100.0%–100.0%]	0.28 [0.25–0.30]
Solar 2107	24	98.3% [97.9%–98.6%]	99.7% [99.6%–99.8%]	0.53 [0.52–0.55]
Solar 9068	24	98.7% [98.1%–99.4%]	99.7% [99.5%–99.9%]	0.02 [0.00–0.05]

- With 4–6 machines all at 100%, the bootstrap shows no spread: a limit of having 4–6 machines, not proof of certainty. With 6 perfect machines the one-sided 95% lower bound on the share of fully covered machines is 61%.
- Hill of Towie (21 turbines) and the solar plants (24 inverters or combiners) give the tightest intervals.

14. Summary

- **The guarantee held:** 0 over-claims when the range was right, all 7 datasets.
- **Best law allowance (13 rules × 7 datasets):** K-S3 with a published limit and machine history (proves 0.93–0.94); K-S1 or Z otherwise; single-unit rules fail beyond the limit.
- **Real neural surrogates (FNOs):** valid 99–100%; on diffusion proves 63% of the error when it speaks; a generic Lipschitz law proves nothing; silent where in-range errors have a heavy tail.
- **Anchor:** put it where the model was trained (b small). Elsewhere the bound goes to 0.
- **Law allowance does most of the work;** the anchor allowance closes the last gap.
- **Ranking:** works, but the raw defect ranks just as well. The allowances buy the guarantee.

Limits

- A lower bound: S = 0 means “no proven error”, not “accurate”.
- Bands are 2–4× the real law error below the limit and 6–14× beyond it.
- A derated or curtailed machine breaks the published limit (Kelmarsh 2016).
- One seed per model type; units share machines; surrogates extrapolate poorly, which flatters “error proven”.

References

[1] Milanese, M. & Novara, C. (2004). Set membership identification of nonlinear systems. *Automatica* 40(6).

[2] Milanese, M. & Novara, C. (2011). Unified set membership theory for identification, prediction and filtering of nonlinear systems. *Automatica* 47. doi:10.1016/j.automatica.2011.03.013

[3] Calliess, J.-P. (2016). Lazily adapted constant kinky inference for nonparametric regression and model-reference adaptive control. arXiv:1701.00178.

[4] Sullivan, T. J., McKerns, M., Meyer, D., Theil, F., Owhadi, H. & Ortiz, M. (2012). Optimal uncertainty quantification for legacy data observations of Lipschitz functions. arXiv:1202.1928.

[5] Groeneboom, P. & Jongbloed, G. (2014). *Nonparametric Estimation under Shape Constraints.* Cambridge University Press.

[6] Vovk, V., Gammerman, A. & Shafer, G. (2005). *Algorithmic Learning in a Random World.* Springer.

[7] Papadopoulos, H., Proedrou, K., Vovk, V. & Gammerman, A. (2002). Inductive confidence machines for regression. *ECML*.

[8] Lei, J., G’Sell, M., Rinaldo, A., Tibshirani, R. J. & Wasserman, L. (2018). Distribution-free predictive inference for regression. *JASA* 113(523).

[9] Barber, R. F., Candès, E. J., Ramdas, A. & Tibshirani, R. J. (2021). Predictive inference with the jackknife+. *Annals of Statistics* 49(1).

[10] Liu, Z. et al. (2024). KAN: Kolmogorov–Arnold Networks. arXiv:2404.19756.

[11] Burton, T., Jenkins, N., Sharpe, D. & Bossanyi, E. (2011). *Wind Energy Handbook*, 2nd ed. Wiley.

[12] Manwell, J. F., McGowan, J. G. & Rogers, A. L. (2009). *Wind Energy Explained*, 2nd ed. Wiley.

[13] King, D. L., Boyson, W. E. & Kratochvil, J. A. (2004). *Photovoltaic Array Performance Model.* Sandia SAND2004-3535.

[14] King, D. L., Gonzalez, S., Galbraith, G. M. & Boyson, W. E. (2007). *Performance Model for Grid-Connected Photovoltaic Inverters.* Sandia SAND2007-5036.

[15] Zakharov, V. E. & Filonenko, N. N. (1967). Weak turbulence of capillary waves. *J. Appl. Mech. Tech. Phys.* 8(5).

[16] Orosco, J., Connacher, W. & Friend, J. (2023). Data from: Identification of weakly- to strongly-turbulent three-wave processes in a micro-scale system. UC San Diego Library. doi:10.6075/J0WW7HVJ

[17] Rozza, G., Huynh, D. B. P. & Patera, A. T. (2008). Reduced basis approximation and a posteriori error estimation for affinely parametrized elliptic coercive PDEs. *Arch. Comput. Methods Eng.* 15(3).

[18] Takamoto, M., Praditia, T., Leiteritz, R., MacKinlay, D., Alesiani, F., Pflüger, D. & Niepert, M. (2022). PDEBench: an extensive benchmark for scientific machine learning. *NeurIPS Datasets and Benchmarks*.

[19] Li, Z., Kovachki, N., Azizzadenesheli, K., Liu, B., Bhattacharya, K., Stuart, A. & Anandkumar, A. (2021). Fourier neural operator for parametric partial differential equations. *ICLR*.

[20] Data: Kelmarsh and Penmanshiel (Cubico, Zenodo 16807551, 16807304); Hill of Towie (RES, Zenodo 22662930); La Haute Borne (ENGIE, via NREL OpenOA); PVDAQ 9068 and 2107 (NREL, OEDI 2023 Solar Data Prize).

Journal details for [1], [5], [7], [11]–[17] are given from memory: verify volume, pages and edition before submission.